

数学

爱豆
爱做题

WOW!



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参考答案

第1讲 三角形

1. 【答案】 $90^\circ - 2\alpha$ 【解析】 $\because \angle CBD = \alpha$,

$$\therefore \angle C = 2\angle CBD = 2\alpha,$$

设 $\angle CAE = \beta$,

$$\therefore \angle BEF = 2\angle CAE = 2\beta,$$

$$\because \angle BAC = 90^\circ,$$

$$\therefore \angle BAE = \angle BAC - \angle CAE = 90^\circ - \beta,$$

 $\therefore \angle BEA$ 为 $\triangle ACE$ 的一个外角,

$$\therefore \angle BEA = \angle C + \angle CAE = 2\alpha + \beta,$$

$$\therefore AB = BE,$$

$$\therefore \angle BAE = \angle BEA,$$

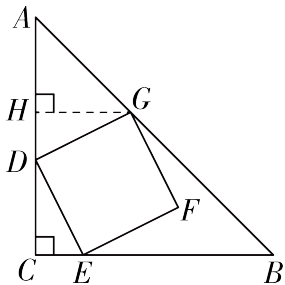
$$\therefore 90^\circ - \beta = 2\alpha + \beta,$$

$$\therefore 2\beta = 90^\circ - 2\alpha,$$

$$\therefore \angle BEF = 2\beta = 90^\circ - 2\alpha,$$

故答案为: $90^\circ - 2\alpha$.2. 【答案】 $3\sqrt{2}$ 【解析】过点 G 作 $GH \perp AC$, 则

$$\angle AHG = \angle GHD = 90^\circ,$$



$$\therefore \angle DGH + \angle HDG = 90^\circ,$$

$$\because \angle ACB = 90^\circ, AC = BC = 5,$$

$$\therefore AB = 5\sqrt{2}, \angle A = \angle B = 45^\circ,$$

$$\therefore \angle AGH = 45^\circ = \angle A,$$

$$\therefore AH = HG,$$

设 $AH = HG = x$, 则 $CH = AC - AH = 5 - x$, $\therefore$ 四边形 $DEFG$ 是正方形,

$$\therefore DG = DE, \angle GDE = 90^\circ,$$

$$\therefore \angle HDG + \angle CDE = 90^\circ,$$

$$\therefore \angle HGD = \angle CDE,$$

$$\because \angle C = \angle GHD = 90^\circ,$$

$$\therefore \triangle GHD \cong \triangle DCE,$$

$$\therefore CD = GH = x,$$

$$\therefore DH = CH - CD = 5 - 2x,$$

Rt $\triangle GHD$ 中, 由勾股定理, 得: $GD^2 = DH^2 + GH^2$,

$$\therefore (\sqrt{5})^2 = (5 - 2x)^2 + x^2, \text{解得: } x = 2,$$

$$\therefore AH = 2, CH = 3,$$

$$\because \angle C = \angle AHG = 90^\circ,$$

$$\therefore HG \parallel BC,$$

$$\therefore \frac{AG}{BG} = \frac{AH}{CH} = \frac{2}{3},$$

$$\therefore BG = \frac{3}{5}AB = \frac{3}{5} \times 5\sqrt{2} = 3\sqrt{2}.$$

故答案为: $3\sqrt{2}$.

3. 【答案】6

【解析】如图,

 $\therefore$ 线段 OP 绕点 O 逆时针旋转 60° 得到线段 OD , 要使点 D 恰好落在 AC 边上,

$$\therefore OP = OD, \angle POD = 60^\circ,$$

$$\therefore \angle 1 + \angle 2 = 120^\circ,$$

 $\therefore \triangle ABC$ 为等边三角形,

$$\therefore \angle A = \angle B = 60^\circ,$$

$$\therefore \angle 2 + \angle 3 = 120^\circ,$$

$$\therefore \angle 1 = \angle 3,$$

在 $\triangle AOD$ 和 $\triangle BPO$ 中

$$\begin{cases} \angle A = \angle B \\ \angle 1 = \angle 3 \\ PO = OD \end{cases},$$

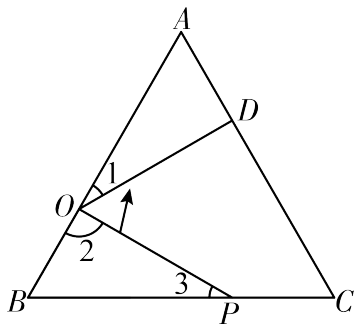
$$\therefore \triangle AOD \cong \triangle BPO,$$

$$\therefore BP = AO,$$

$$\text{而 } AO = AB - OB = 9 - 3 = 6,$$

$$\therefore BP = 6.$$

故答案为 6.



4. 【答案】D

【解析】 $\because$ 四边形 $ABCD$ 是正方形,

$$\therefore AB = AD, \angle ABE = \angle ADF = 90^\circ,$$

在 $\triangle ABE$ 和 $\triangle ADF$ 中,

$$\begin{cases} AB = AD \\ \angle ABE = \angle ADF \\ BE = DF \end{cases},$$

$$\therefore \triangle ABE \cong \triangle ADF \text{ (SAS)},$$

$$\therefore AE = AF,$$

$$\therefore AM \text{ 平分 } \angle EAF,$$

$$\therefore \angle EAM = \angle FAM,$$

在 $\triangle AEM$ 和 $\triangle AFM$ 中,

$$\begin{cases} AE = AF \\ \angle EAM = \angle FAM \\ AM = AM \end{cases},$$

$$\therefore \triangle AEM \cong \triangle AFM \text{ (SAS)},$$

$$\therefore EM = FM,$$

$\because$ 四边形 $ABCD$ 是正方形,

$$\therefore BC = CD = 4, \angle BCD = 90^\circ,$$

$$\text{设 } DM = x, \text{ 则 } MC = CD - DM = 4 - x,$$

$$CE = BC - BE = 4 - 1 = 3, EM = FM =$$

$$FD + DM = 1 + x,$$

在 $\text{Rt}\triangle MCE$ 中, 根据勾股定理, 得 $EM^2 =$

$$MC^2 + CE^2, \text{ 即 } (1+x)^2 = (4-x)^2 + 3^2,$$

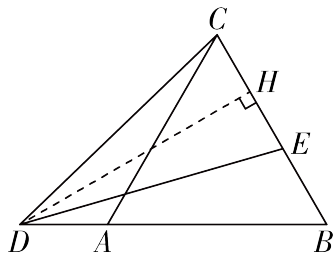
$$\text{解得 } x = \frac{12}{5},$$

$$\text{即 } DM = \frac{12}{5}.$$

故选 D.

5. 【答案】 $\frac{10}{3}\sqrt{6}$

【解析】作 $DH \perp BC$ 于 H , 则: $\angle DHB = 90^\circ$,



$$\because DE = DC,$$

$$\therefore CH = EH = \frac{1}{2}CE = \sqrt{6},$$

$$\because \angle ABC \text{ 为等边三角形},$$

$$\therefore \angle B = 60^\circ,$$

$$\therefore \angle BDH = 30^\circ.$$

$$\therefore BD = 2BH = 2(BE + HE),$$

$$\therefore BD + BE = 6\sqrt{6},$$

$$\therefore 2(BE + \sqrt{6}) + BE = 6\sqrt{6}.$$

$$\therefore BE = \frac{4}{3}\sqrt{6},$$

$$\therefore BC = CE + BE = \frac{10}{3}\sqrt{6}. \text{ 即等边三角形的边长为 } \frac{10}{3}\sqrt{6}.$$

$$\text{故答案为: } \frac{10}{3}\sqrt{6}.$$

6. 【答案】 6

【解析】 $\because$ 在 $\triangle ABC$ 中, 点 D, E 分别是边 AB, BC 的中点, $AC = 4$,

$$\therefore DE = \frac{1}{2}AC = 2,$$

$$\because \angle BFC = 90^\circ, BC = 8,$$

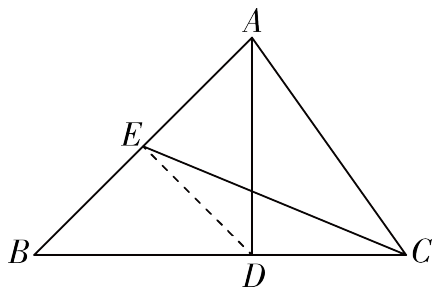
$$\therefore EF = \frac{1}{2}BC = 4,$$

$$\therefore DF = DE + EF = 6.$$

故答案为: 6.

7. 【答案】 $\frac{\sqrt{6} + \sqrt{3}}{3}a$

【解析】如图, 连接 DE ,



$\because AD$ 是 BC 边上的高线,

$\therefore AD \perp BC$,

$\because CE$ 是 AB 边上的中线,

$\therefore$ 点 E 为 AB 中点,

$$\therefore CD = AE = BE = DE = \frac{1}{2}AB,$$

$$\therefore \angle CED = \angle ECD,$$

$$\therefore \angle BDE = \angle CED + \angle ECD = 2\angle BCE,$$

$$\therefore \angle BAD = 2\angle BCE,$$

$$\therefore \angle BDE = \angle BAD,$$

$$\text{又 } \because \angle B = \angle B,$$

$$\therefore \triangle BDE \sim \triangle BAD,$$

$$\therefore \angle BED = \angle BDA = 90^\circ,$$

$\therefore \triangle BED$ 为等腰直角三角形,

$$\therefore BD = \sqrt{2}DE,$$

设 $CD = AE = BE = DE = x$, 则 $BD = \sqrt{2}x$,

$$BC = \sqrt{2}x + x = (\sqrt{2} + 1)x, AB = 2x,$$

$$\therefore AD = \sqrt{AB^2 - BD^2} = \sqrt{(2x)^2 - (\sqrt{2}x)^2} = \sqrt{2}x,$$

$$\therefore AD^2 + CD^2 = AC^2, \text{ 即 } (\sqrt{2}x)^2 + x^2 = a^2,$$

$$\text{解得: } x = \frac{\sqrt{3}a}{3},$$

$$\therefore BC = (\sqrt{2} + 1) \times \frac{\sqrt{3}a}{3} = \frac{(\sqrt{6} + \sqrt{3})a}{3},$$

$$\text{故答案为: } \frac{(\sqrt{6} + \sqrt{3})a}{3}.$$

8. 【答案】 A

【解析】 $\because \angle ACB = 90^\circ, \angle CAB = 30^\circ$,

$$\therefore AB = 2BC, AC = \sqrt{3}BC,$$

设 $BC = x$, 则 $AB = 2x, AC = \sqrt{3}x$,

$\because AD$ 平分 $\angle CAB, \angle ACB = 90^\circ$,

$\therefore$ 点 D 到 AC, AB 的距离相等, 均为 CD

的长, $\angle CAD = \angle BAD$,

$$\therefore \frac{S_{\triangle ACD}}{S_{\triangle ABD}} = \frac{\frac{1}{2}AC \cdot CD}{\frac{1}{2}AB \cdot CD} = \frac{CD}{BD},$$

$$\therefore \frac{CD}{BD} = \frac{AC}{AB} = \frac{\sqrt{3}}{2},$$

$$\therefore CD = \frac{\sqrt{3}}{2 + \sqrt{3}}BC = (2\sqrt{3} - 3)x,$$

$$\therefore AD = \sqrt{AC^2 + CD^2} = (3\sqrt{2} - \sqrt{6})x,$$

$$\because BE \perp AD, \angle CAD = \angle BAD,$$

$$\begin{aligned} \therefore \sin \angle CAD &= \sin \angle BAD, \\ \therefore \frac{CD}{AD} &= \frac{BE}{AB}, \text{ 即 } \frac{BE}{2x} = \frac{2\sqrt{3}-3}{3\sqrt{2}-\sqrt{6}}, \\ \therefore BE &= \frac{\sqrt{6}-\sqrt{2}}{2}x, \\ \therefore \frac{AD}{BE} &= \frac{(3\sqrt{2}-\sqrt{6})x}{\frac{\sqrt{6}-\sqrt{2}}{2}x} = 2\sqrt{3}. \end{aligned}$$

故选:A.

第2讲 相似

1. 【答案】 $\frac{2}{9}$

【解析】 $\because G_1, G_2$ 是 $\triangle APB, \triangle APC$ 的重心,

$$\therefore \frac{AG_1}{AE} = \frac{AG_2}{AF} = \frac{2}{3},$$

$$\therefore \angle G_1AG_2 = \angle EAF,$$

$$\therefore \triangle AG_1G_2 \sim \triangle AEF,$$

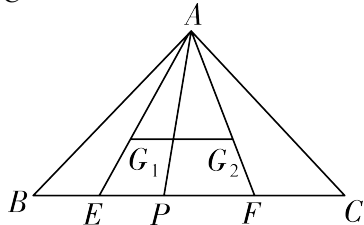
$$\frac{S_{\triangle AG_1G_2}}{S_{\triangle AEF}} = \frac{4}{9},$$

$\because E, F$ 分别是 BP, CP 的中点,

$$\therefore \frac{S_{\triangle AEP}}{S_{\triangle ABP}} = \frac{1}{2}, \frac{S_{\triangle APF}}{S_{\triangle APC}} = \frac{1}{2},$$

$$\therefore \frac{S_{\triangle AEF}}{S_{\triangle ABC}} = \frac{1}{2},$$

$$\therefore \frac{S_{\triangle AG_1G_2}}{S_{\triangle ABC}} = \frac{4}{9} \times \frac{1}{2} = \frac{2}{9},$$



故答案为: $\frac{2}{9}$.

2. 【答案】3

【解析】如图,

$\because S_{\triangle ABC} = 16, S_{\triangle A'EF} = 9$, 且 AD 为 BC 边的中线,

$$\begin{aligned} \therefore S_{\triangle A'DE} &= \frac{1}{2} S_{\triangle A'EF} = 4.5, S_{\triangle ABD} = \\ &= \frac{1}{2} S_{\triangle ABC} = 8, \end{aligned}$$

$\therefore$ 将 $\triangle ABC$ 沿 BC 边上的中线 AD 平移得到 $\triangle A'B'C'$,

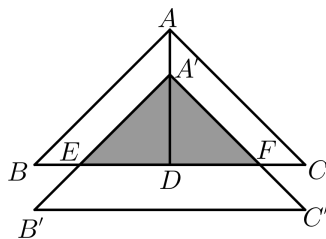
$$\therefore A'E \parallel AB,$$

$$\therefore \triangle DA'E \sim \triangle DAB,$$

$$\text{则 } \left(\frac{A'D}{AD} \right)^2 = \frac{S_{\triangle A'DE}}{S_{\triangle ADB}}, \text{ 即 } \left(\frac{A'D}{A'D+1} \right)^2 = \frac{4.5}{8},$$

解得 $A'D = 3$ 或 $A'D = -\frac{3}{7}$ (舍),

故答案为 3.



3. 【答案】 $\sqrt{m^2+n^2}$

【解析】 $\because \triangle ABC$ 是等边三角形,

$$\therefore \angle A = \angle B = \angle C = 60^\circ,$$

$\therefore$ 折叠 $\triangle BDE$ 得到 $\triangle FDE$,

$$\therefore \triangle BDE \cong \triangle FDE,$$

$$\therefore S_{\triangle BDE} = S_{\triangle FDE}, \angle F = \angle B = 60^\circ = \angle A = \angle C,$$

$\therefore DE$ 平分等边 $\triangle ABC$ 的面积,

$$\therefore S_{\text{四边形}ACED} = S_{\triangle BDE} = S_{\triangle FDE},$$

$$\therefore S_{\triangle FHG} = S_{\triangle ADG} + S_{\triangle CHE},$$

又 $\because \angle AGD = \angle FGH, \angle CHE = \angle FHG$,

$$\therefore \triangle ADG \sim \triangle FHG, \triangle CHE \sim \triangle FHG,$$

$$\begin{aligned} \therefore \frac{S_{\triangle ADG}}{S_{\triangle FHG}} &= \left(\frac{DG}{GH} \right)^2 = \frac{m^2}{GH^2}, \frac{S_{\triangle CHE}}{S_{\triangle FHG}} = \\ &= \left(\frac{EH}{GH} \right)^2 = \frac{n^2}{GH^2}, \end{aligned}$$

$$\begin{aligned} \therefore \frac{S_{\triangle ADG}}{S_{\triangle FHG}} + \frac{S_{\triangle CHE}}{S_{\triangle FHG}} &= \\ \frac{m^2 + n^2}{GH^2} &= \frac{S_{\triangle ADG} + S_{\triangle CHE}}{S_{\triangle FHG}} = 1, \\ \therefore GH^2 &= m^2 + n^2, \end{aligned}$$

解得 $GH = \sqrt{m^2 + n^2}$ 或 $GH = -\sqrt{m^2 + n^2}$

(不符合题意,舍去),

故答案为: $\sqrt{m^2 + n^2}$.

4. 【答案】5

【解析】 $\because AE \parallel BC$,

$\therefore \triangle AEG \sim \triangle BFG$,

$\therefore BG : GA = 3 : 1 = BF : AE$,

$\therefore D$ 为 AC 的中点,

$\therefore AD = CD$.

$\therefore AE \parallel CF$,

$\therefore \angle EAD = \angle FCD, \angle AED = \angle DFC$.

$\therefore \triangle EAD \cong \triangle FCD$.

$\therefore AE = CF$,

$\therefore BF : AE = (CF + BC) : AE = 3 : 1$,

$\therefore (AE + 10) : AE = 3 : 1$,

解得: $AE = 5$.

因此正确答案为:5.

5. 【答案】B

【解析】 $\because$ 四边形 $ABCD$ 是正方形, $AF = 2$,

$FB = 1$,

$\therefore AD = BC = AB = AF + FB = 2 + 1 = 3$,

$AD \parallel CB, AD \perp AB, CB \perp AB$,

$\therefore EF \perp AB$,

$\therefore AD \parallel EF \parallel BC$,

$\therefore \frac{DE}{EM} = \frac{AF}{FB} = 2, \triangle ADE \sim \triangle CME$,

$\therefore \frac{AD}{CM} = \frac{DE}{EM} = 2$,

则 $CM = \frac{1}{2}AD = \frac{3}{2}$,

$\therefore MB = 3 - CM = \frac{3}{2}$,

$\therefore BC \parallel AD$,

$\therefore \triangle GMB \sim \triangle GDA$,

$\therefore \frac{BG}{AG} = \frac{MB}{DA} = \frac{\frac{3}{2}}{3} = \frac{1}{2}$

$\therefore BG = AB = 3$,

在 $\text{Rt}\triangle BGM$ 中, $MG = \sqrt{MB^2 + BG^2} =$

$\sqrt{\left(\frac{3}{2}\right)^2 + 3^2} = \frac{3\sqrt{5}}{2}$.

故选 B.

6. 【答案】B

【解析】 $\because$ 矩形 $ABCD$, E, F 是 BC 边上的三等分点, $AB = 8, BC = 12$,

$\therefore AD = BC = 12, CD = AB = 8, AD \parallel BC$,

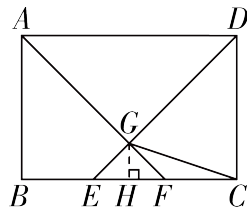
$BE = EF = FC = 4, EC = 8$,

$\therefore \triangle AGD \sim \triangle FGE$,

$\therefore \frac{EG}{DG} = \frac{EF}{AD} = \frac{4}{12} = \frac{1}{3}$,

$\therefore \frac{EG}{ED} = \frac{1}{4}$,

过点 G 作 $GH \perp BC$,



则 $GH \parallel CD$,

$\therefore \triangle GHE \sim \triangle DCE$,

$\therefore \frac{EH}{EC} = \frac{GH}{CD} = \frac{EG}{DE} = \frac{1}{4}$,

$\therefore EH = \frac{1}{4}EC = \frac{1}{4} \times 8 = 2, GH = \frac{1}{4}CD = \frac{1}{4} \times 8 = 2$,

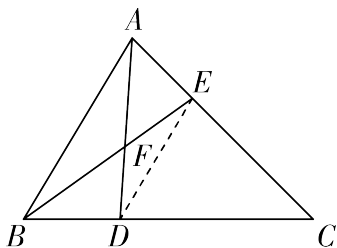
$\therefore CH = CE - EH = 8 - 2 = 6$,

$$\therefore \tan \angle GCF = \frac{GH}{CH} = \frac{2}{6} = \frac{1}{3}.$$

故选:B.

7. 【答案】 $\frac{4}{3}$

【解析】如图,连接 DE ,



$$\because CE = 2AE,$$

$$\therefore \frac{S_{\triangle ABE}}{S_{\triangle CBE}} = \frac{AE}{CE} = \frac{1}{2},$$

$$\text{即 } S_{\triangle CBE} = 2S_{\triangle ABE},$$

$$\text{又 } \because S_{\triangle ABE} + S_{\triangle CBE} = 10,$$

$$\therefore S_{\triangle ABE} = \frac{10}{3},$$

$$\text{由题意知, } \frac{CD}{BC} = \frac{CD}{BD+CD} = \frac{2}{3}, \quad \frac{CE}{AC} =$$

$$\frac{CE}{AE+CE} = \frac{2}{3},$$

$$\therefore \frac{CD}{BC} = \frac{CE}{AC},$$

$$\therefore \angle DCE = \angle BCA,$$

$$\therefore \triangle DCE \sim \triangle BCA,$$

$$\therefore \frac{DE}{AB} = \frac{CD}{BC} = \frac{2}{3}, \angle CDE = \angle CBA,$$

$$\therefore DE \parallel AB,$$

$$\therefore \angle EDF = \angle BAF, \angle DEF = \angle ABF,$$

$$\therefore \triangle DEF \sim \triangle ABF,$$

$$\therefore \frac{EF}{BF} = \frac{DE}{AB} = \frac{2}{3},$$

$$\therefore \frac{S_{\triangle AEF}}{S_{\triangle ABF}} = \frac{EF}{BF} = \frac{2}{3}, \text{即 } S_{\triangle ABF} = \frac{3}{2}S_{\triangle AEF},$$

$$\text{又 } \because S_{\triangle AEF} + S_{\triangle ABF} = S_{\triangle ABE} = \frac{10}{3},$$

$$\therefore S_{\triangle AEF} = \frac{4}{3}.$$

故答案为: $\frac{4}{3}$.

8. 【答案】 $\frac{3\sqrt{3}-\sqrt{7}}{2}$

【解析】Rt $\triangle ABC$ 和 Rt $\triangle CDE$ 中, $\angle ACB = \angle DCE = 90^\circ$, $\angle CAB = \angle CDE = 30^\circ$, $BC = 3$, $CE = 2$,

$$\therefore \frac{AC}{BC} = \frac{CD}{EC} = \sqrt{3}, \quad AB = 2BC = 6, \quad DE = 2CE = 4,$$

$$\because \angle ACB = \angle DCE = 90^\circ,$$

$$\therefore \angle ACD = \angle BCE,$$

$$\therefore \triangle ACD \sim \triangle BCE,$$

$$\therefore \frac{AD}{BE} = \frac{AC}{CB} = \sqrt{3}, \angle A = \angle CBE,$$

$$\because \angle A + \angle ABC = 90^\circ,$$

$$\therefore \angle CBE + \angle ABC = 90^\circ,$$

$$\therefore \angle DBE = 90^\circ,$$

$$\text{设 } BE = x, \text{ 则 } AD = \sqrt{3}x,$$

$$\text{在 Rt}\triangle DBE \text{ 中, } DE^2 = BD^2 + BE^2,$$

$$\therefore (6 - \sqrt{3}x)^2 + x^2 = 4^2,$$

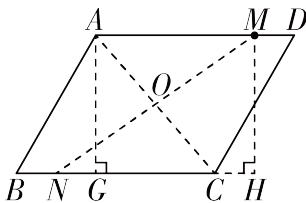
$$\therefore x = \frac{3\sqrt{3}-\sqrt{7}}{2} \text{ 或 } x = \frac{3\sqrt{3}+\sqrt{7}}{2} \text{ (不合题意, 舍去),}$$

$$\therefore BE = \frac{3\sqrt{3}-\sqrt{7}}{2}.$$

第3讲 四边形

1. 【答案】 $2\sqrt{37}$

【解析】如图,取 AC 中点 O ,连接 MO 并延长交 BC 于点 N ,过 A 作 $AG \perp BC$ 于点 G ,过 M 作 $MH \perp BC$,交 BC 的延长线于点 H ,



$$\because \angle B = 60^\circ,$$

$$\therefore \angle BAG = 90^\circ - \angle B = 30^\circ,$$

$$\therefore BG = \frac{1}{2}AB = 4,$$

$$\therefore AG = \sqrt{AB^2 - BG^2} = \sqrt{8^2 - 4^2} = 4\sqrt{3},$$

$\because MN$ 平分四边形 $ABCD$ 的面积,

$\therefore MN$ 经过平行四边形 $ABCD$ 的中心 O ,

$\because$ 在平行四边形 $ABCD$ 中, $AB = 8$,

$$BC = 10,$$

$$\therefore AD = BC = 10, AD \parallel BC,$$

$$\therefore \angle AMO = \angle CNO, \angle MAO = \angle NCO,$$

又 $\because AO = CO$,

$$\therefore \triangle AOM \cong \triangle CON \text{ (AAS)},$$

$$\therefore AM = CN,$$

$$\therefore AM = 8,$$

$$\therefore CN = 8,$$

$$\therefore BN = BC - CN = 2,$$

$$\because AG \perp BC, MH \perp BC,$$

$$\therefore AG \parallel MH,$$

又 $\because AM \parallel GH$,

$\therefore$ 四边形 $AGHM$ 是平行四边形,

$$\therefore GH = AM = 8, MH = AG = 4\sqrt{3},$$

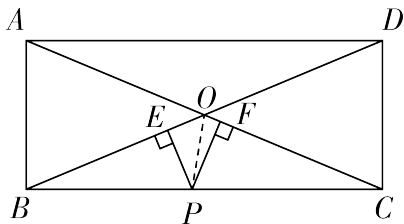
$$\therefore NH = BG + GH - BN = 4 + 8 - 2 = 10,$$

$$\therefore MN = \sqrt{NH^2 + MH^2} = \sqrt{(10)^2 + (4\sqrt{3})^2} = 2\sqrt{37},$$

故答案为: $2\sqrt{37}$.

2. 【答案】 $\frac{120}{13}$

【解析】设 AC 与 BD 相交于点 O , 连接 OP , 如图所示:



$\because$ 四边形 $ABCD$ 是矩形,

$$\therefore \angle ABC = 90^\circ, OA = OC = \frac{1}{2}AC, OB = OD = \frac{1}{2}BD, AC = BD,$$

$$\therefore OB = OC,$$

$$\because AB = 10, BC = 24,$$

$$\therefore AC = \sqrt{AB^2 + BC^2} = \sqrt{10^2 + 24^2} = 26,$$

$$\therefore OB = OC = \frac{1}{2} \times 26 = 13,$$

$$\because S_{\triangle BOC} = S_{\triangle BOP} + S_{\triangle COP},$$

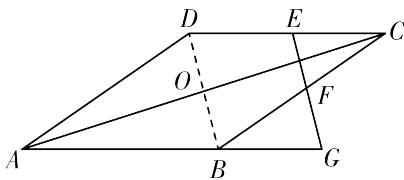
$$\therefore \frac{1}{2} \times \frac{1}{2} \times 10 \times 24 = \frac{1}{2} \times 13 \cdot PE + \frac{1}{2} \times 13 \cdot PF,$$

$$\therefore PE + PF = \frac{120}{13}.$$

故答案为: $\frac{120}{13}$.

3. 【答案】 B

【解析】连接 BD , 交 AC 于点 O , 如图:



$\because$ 菱形 $ABCD$ 的边长为 13, 点 E 、 F 分别是边 CD 、 BC 的中点,

$$\therefore AB \parallel CD, AB = BC = CD = DA = 13,$$

$$EF \parallel BD,$$

$$\because AC、BD \text{ 是菱形的对角线}, AC = 24,$$

$$\therefore AC \perp BD, AO = CO = 12, OB = OD,$$

$$\text{又 } \because AB \parallel CD, EF \parallel BD,$$

$$\therefore DE \parallel BG, BD \parallel EG,$$

∴ 四边形 $BDEG$ 是平行四边形,

$$\therefore BD = EG,$$

在 $\triangle COD$ 中, $\because OC \perp OD, CD = 13, CO = 12,$

$$\therefore OD = \sqrt{13^2 - 12^2} = 5,$$

$$\therefore BD = 2OD = 10,$$

$$\therefore EG = BD = 10.$$

故选 B.

4. 【答案】 $30^\circ \quad 9 - 3\sqrt{3}$

【解析】∵ 四边形 $ABCD$ 是正方形,

$$\therefore AD = CD, \angle ADC = 90^\circ,$$

$$\angle ADB = \angle CDB = 45^\circ,$$

$$\because DF = DF,$$

$$\therefore \triangle ADF \cong \triangle CDF (\text{SAS}),$$

$$\therefore \angle DAF = \angle DCF,$$

$$\because EC = EF,$$

$$\therefore \angle ECF = \angle EFC,$$

$$\therefore \angle AED = 2\angle ECF, \text{ 则 } \angle AED = 2\angle EAD,$$

在 $\text{Rt}\triangle ADE$ 中, $\angle EAD + \angle AED = 90^\circ,$

$$\therefore \angle EAD + 2\angle EAD = 90^\circ,$$

$$\therefore \angle EAD = 30^\circ;$$

在 $\text{Rt}\triangle ADE$ 中, $\angle EAD = 30^\circ, AD = 3\sqrt{3},$

$$\therefore DE = 3, AE = 6,$$

$$\therefore EC = EF = CD - DE = 3\sqrt{3} - 3,$$

$$\therefore AF = AE - EF = 6 - (3\sqrt{3} - 3) = 9 - 3\sqrt{3}.$$

故答案为: $30^\circ; 9 - 3\sqrt{3}.$

5. 【答案】C

【解析】因为 $DE \parallel CA, DF \parallel BA$, 所以四边形 $AEDF$ 是平行四边形, 故 ① 正确.

因为 $\angle BAC = 90^\circ$, 四边形 $AEDF$ 是平行四边形, 所以四边形 $AEDF$ 是矩形, 故 ② 正确.

因为 AD 平分 $\angle BAC$, 所以 $AE = DE$, 又因为四边形 $AEDF$ 是平行四边形, 所以是菱形, 故 ③ 正确.

如果 $AD \perp BC$ 且 $AB = AC$ 不能判定四边形 $AEDF$ 是正方形, 故 ④ 错误.

故选: C.

6. 【答案】A

【解析】在正方形 $ABCD$ 中, $BC = CD,$
 $\angle BCD = \angle ADC = 90^\circ,$

$$\because CE = DH,$$

$$\therefore \triangle BCE \cong \triangle CDH (\text{SAS}),$$

$$\therefore \angle CBE = \angle DCH, BE = CH, \text{ 故 ① 正确;}$$

$$\because \angle DCH + \angle FCB = \angle DCB = 90^\circ,$$

$$\therefore \angle CBF + \angle FCB = 90^\circ,$$

$$\therefore \angle BFC = 90^\circ,$$

$$\therefore CH \perp BE, \text{ 故 ② 正确;}$$

$$\because G \text{ 在正方形的对角线 } BD \text{ 上,}$$

$$\therefore G \text{ 到 } AD, CD \text{ 的距离相等,}$$

$$\therefore CE = DH,$$

$$\therefore S_{\triangle GCE} = S_{\triangle GDH}, \text{ 故 ③ 正确;}$$

设正方形 $ABCD$ 的边长为 $4a,$

$$\because BC = CD = 4a, \text{ 当 } E \text{ 是 } CD \text{ 的中点时,}$$

$$EC = HD = 2a,$$

$$\text{由勾股定理得: } BE = \sqrt{BC^2 + CE^2}$$

$$= \sqrt{(4a)^2 + (2a)^2} = 2\sqrt{5}a = CH,$$

$$\therefore \angle HDG = \angle CBG = 45^\circ, \angle HGD = \angle CGB,$$

$$\therefore \triangle HGD \sim \triangle CGB,$$

$$\therefore \frac{HG}{CG} = \frac{DH}{BC} = \frac{1}{2},$$

$$\therefore GC = \frac{2}{3}CH,$$

$$\therefore \angle BEC = \angle CEF, \angle ECB = \angle EFC = 90^\circ,$$

$$\therefore \triangle ECB \sim \triangle EFC,$$

$$\therefore \frac{CE}{EF} = \frac{BE}{CE},$$

$$\therefore \frac{2a}{EF} = \frac{2\sqrt{5}a}{2a},$$

$$\therefore EF = \frac{2\sqrt{5}a}{5},$$

$$\therefore CF = \sqrt{CE^2 - EF^2} = \sqrt{(2a)^2 - \left(\frac{2\sqrt{5}a}{5}\right)^2} = \frac{4\sqrt{5}a}{5},$$

$$\therefore GC = \frac{2}{3}CH = \frac{2}{3} \times 2\sqrt{5}a = \frac{4\sqrt{5}a}{3},$$

$$\therefore GF = CG - CF = \frac{4\sqrt{5}a}{3} - \frac{4\sqrt{5}a}{5} = \frac{8\sqrt{5}a}{15},$$

$$\therefore GE = \sqrt{GF^2 + EF^2} = \sqrt{\left(\frac{8\sqrt{5}a}{15}\right)^2 + \left(\frac{2\sqrt{5}a}{5}\right)^2} = \frac{2\sqrt{5}a}{3},$$

$$\therefore \frac{GF}{GE} = \frac{8\sqrt{5}a}{15} \times \frac{3}{2\sqrt{5}a} = \frac{4}{5},$$

$\therefore$ 当 E 是 CD 的中点时, $\frac{GF}{GE} = \frac{4}{5}$, 故 ④ 正确,

$$\text{当 } EC = 2DE \text{ 时, } \frac{CE}{CD} = \frac{2}{3},$$

$$\therefore DH = CE, DC = BC,$$

$$\therefore \frac{DH}{BC} = \frac{CE}{CD} = \frac{2}{3},$$

$$\therefore \triangle HGD \sim \triangle CGB,$$

$$\therefore \frac{S_{\triangle HGD}}{S_{\triangle CGB}} = \left(\frac{DH}{BC}\right)^2 = \frac{4}{9},$$

$\therefore \triangle GDH$ 中 DH 边上的高与 $\triangle DGC$ 中 CD 边上的高相等, $\frac{DH}{BC} = \frac{DH}{CD} = \frac{2}{3},$

$$\therefore \frac{S_{\triangle GDH}}{S_{\triangle DGC}} = \frac{DH}{CD} = \frac{2}{3},$$

设 $S_{\triangle GDH} = 4x$, 则 $S_{\triangle CGB} = 9x, S_{\triangle DGC} = 6x,$

$$\therefore S_{\triangle BCD} = S_{\triangle CGB} + S_{\triangle DGC} = 9x + 6x = 15x,$$

$$\therefore S_{\text{正方形}ABCD} = 2S_{\triangle BCD} = 30x,$$

$$\text{当 } EC = 2DE \text{ 时, } \frac{DE}{CD} = \frac{1}{3},$$

$$\therefore \frac{S_{\triangle DEG}}{S_{\triangle DCG}} = \frac{1}{3},$$

$$\therefore S_{\triangle DEG} = 2x,$$

$$\therefore S_{\text{四边形}DEGH} = S_{\triangle GDH} + S_{\triangle DEG} = 4x + 2x = 6x,$$

$\therefore S_{\text{正方形}ABCD} = 5S_{\text{四边形}DEGH}$, 故 ⑤ 不正确,

综上所述: 正确结论的序号是 ①②③④.

故选: A.

7. 【答案】B

【解析】 $\therefore$ 四边形 $EFGH$ 为正方形,

$$\therefore \angle EGH = 45^\circ, \angle FGH = 90^\circ,$$

$$\therefore OG = GP,$$

$$\therefore \angle GOP = \angle OPG = 67.5^\circ,$$

$$\therefore \angle PBG = 22.5^\circ,$$

$$\text{又 } \therefore \angle DBC = 45^\circ,$$

$$\therefore \angle GBC = 22.5^\circ,$$

$$\therefore \angle PBG = \angle GBC,$$

$$\therefore \angle BGP = \angle BGC = 90^\circ, BG = BG,$$

$$\therefore \triangle BPG \cong \triangle BCG (\text{ASA}),$$

$$\therefore PG = CG.$$

$$\text{设 } OG = PG = CG = x,$$

$$\therefore O \text{ 为 } EG, BD \text{ 的交点,}$$

$$\therefore EG = 2x, FG = \sqrt{2}x,$$

∵ 四个全等的直角三角形拼成“赵爽弦图”，

$$\therefore BF = CG = x,$$

$$\therefore BG = x + \sqrt{2}x,$$

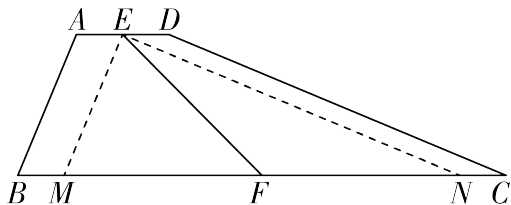
$$\therefore BC^2 = BG^2 + CG^2 = x^2(\sqrt{2}+1)^2 + x^2 = (4+2\sqrt{2})x^2,$$

$$\therefore \frac{S_{\text{正方形}ABCD}}{S_{\text{正方形}EFGH}} = \frac{(4+2\sqrt{2})x^2}{2x^2} = 2 + \sqrt{2},$$

故选:B.

8. 【答案】 $\frac{13}{2}$

【解析】如图，过点 E 作 $EM \parallel AB$ ，交 BC 于点 M ，作 $EN \parallel CD$ ，交 BC 于点 N ，



∵ 四边形 $ABCD$ 为梯形且 $AD \parallel BC$ ，

又 ∵ $EM \parallel AB$ ，

∴ 四边形 $ABME$ 为平行四边形，

$$\therefore AB = EM, \angle B = \angle EMF, BM = AE,$$

∵ $AD \parallel BC, EN \parallel CD$ ，

∴ 四边形 $CDEN$ 为平行四边形，

$$\therefore EN = CD, \angle C = \angle ENF, ED = NC,$$

$$\therefore \angle EMF + \angle ENF = \angle B + \angle C = 90^\circ,$$

$$\therefore \angle MEN = 180^\circ - 90^\circ = 90^\circ,$$

∴ $\triangle EMN$ 为直角三角形，

$$\therefore MN = \sqrt{EM^2 + EN^2}$$

$$= \sqrt{AB^2 + CD^2}$$

$$= \sqrt{5^2 + 12^2}$$

$$= 13.$$

∵ 点 E, F 分别为 AD, BC 的中点，

$$\therefore AE = ED,$$

$$BF = FC,$$

$$\therefore BF - BM = FC - CN,$$

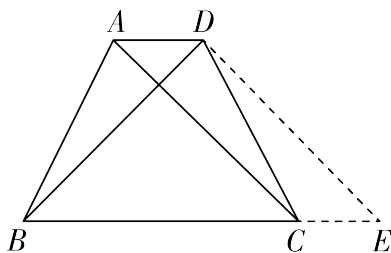
即 $MF = NF$ ，

$$\therefore EF = \frac{1}{2}MN = \frac{13}{2}.$$

故答案为: $\frac{13}{2}$.

9. 【答案】 A

【解析】过 D 作 $DE \parallel CA$ ，交 BC 延长线于 E ，如图所示：



∵ $AD \parallel BC$ ，

∴ 四边形 $ACED$ 是平行四边形，

$$\therefore CE = AD, AC = DE,$$

∵ 四边形 $ABCD$ 是等腰梯形，

$$\therefore \angle DAB = \angle CDA,$$

$$\therefore AD = DA, AB = CD$$

$$\therefore \triangle DAB \cong \triangle ADC (\text{SAS}),$$

$$\therefore AC = BD, \text{即 } BD = DE,$$

$$\therefore AC \perp BD,$$

$$\therefore DE \perp BD,$$

在 $\text{Rt}\triangle BDE$ 中， $BD = DE, AD = a, BC = b$ ，

$$\therefore BE = BC + CE = b + a,$$

$$\therefore AC = DE = \frac{BE}{\sqrt{2}} = \frac{\sqrt{2}}{2}BE = \frac{\sqrt{2}}{2}(a+b), \text{此时 ① 正确;}$$

由 $\triangle DAB \cong \triangle ADC$ ，

$$\therefore S_{\triangle DAB} = S_{\triangle ADC} = S_{\triangle DCE},$$

$$\therefore S_{\text{梯形}ABCD} = S_{\triangle BDE} = \frac{1}{2}BD \cdot AC = \frac{1}{2} \times$$

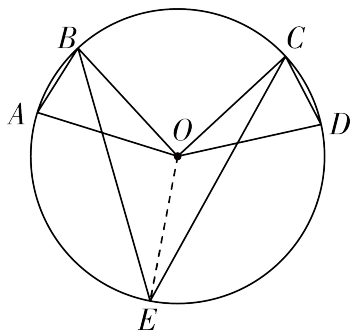
$$\left[\frac{\sqrt{2}}{2}(a+b) \right]^2 = \frac{(a+b)^2}{4}, \text{故 ② 错误.}$$

故选 A.

第4讲 圆

1. 【答案】A

【解析】如图, 连接 OE ,



$$\therefore OA = OB,$$

$$\therefore \angle OBA = \angle A,$$

$$\therefore \angle AOB = 180^\circ - 2\angle A,$$

同理可得: $\angle COD = 180^\circ - 2\angle D$,

$$\therefore \angle A + \angle D = 150^\circ,$$

$$\therefore \angle AOB + \angle COD = 360^\circ - 2(\angle A + \angle D) = 360^\circ - 2 \times 150^\circ = 60^\circ,$$

$$\therefore OB \perp OC,$$

$$\therefore \angle BOC = 90^\circ,$$

$$\therefore \angle AOE + \angle EOD = 360^\circ - (\angle AOB + \angle COD) - \angle BOC = 360^\circ - 60^\circ - 90^\circ = 210^\circ,$$

$$\text{由圆周角定理得: } \angle ABE = \frac{1}{2}\angle AOE,$$

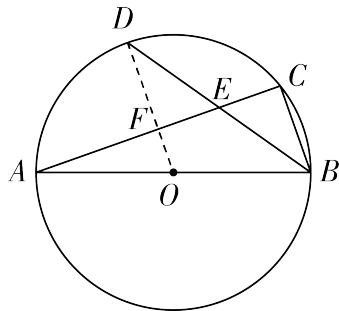
$$\angle DCE = \frac{1}{2}\angle EOD,$$

$$\therefore \angle ABE + \angle DCE = \frac{1}{2}\angle AOE + \frac{1}{2}\angle EOD = \frac{1}{2}(\angle AOE + \angle EOD) = \frac{1}{2} \times 210^\circ = 105^\circ.$$

故选 A.

2. 【答案】 $4\sqrt{2}$

【解析】如下图所示, 连接 OD , 交 AC 于 F ,



$\therefore D$ 是 $\widehat{AC}$ 的中点,

$$\therefore OD \perp AC, AF = CF,$$

$$\therefore \angle DFE = 90^\circ,$$

$$\therefore OA = OB, AF = CF,$$

$$\therefore OF = \frac{1}{2}BC,$$

$\therefore AB$ 是直径,

$$\therefore \angle ACB = 90^\circ,$$

在 $\triangle EFD$ 和 $\triangle ECB$ 中,

$$\begin{cases} \angle DFE = \angle BCE = 90^\circ \\ \angle DEF = \angle BEC \\ DE = BE \end{cases},$$

$$\therefore \triangle EFD \cong \triangle ECB (\text{AAS}),$$

$$\therefore DF = BC,$$

$$\therefore OF = \frac{1}{2}DF,$$

$$\therefore OD = 3,$$

$$\therefore OF = 1, AB = 2OD = 6,$$

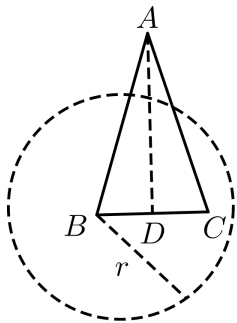
$$\therefore BC = 2,$$

$$\therefore AC = \sqrt{AB^2 - BC^2} = \sqrt{6^2 - 2^2} = 4\sqrt{2}.$$

因此正确答案为: $4\sqrt{2}$.

3. 【答案】 $10 < r < 13$

【解析】如图,过点 A 作 $AD \perp BC$ 于 D ,



$$\because AB = AC, AD \perp BC,$$

$$\therefore BD = \frac{1}{2}BC = \frac{1}{2} \times 10 = 5, \angle ADB = 90^\circ,$$

$$\therefore \cot B = \frac{BD}{AD} = \frac{5}{12}, \text{即 } \frac{5}{AD} = \frac{5}{12}$$

$$\therefore AD = 12,$$

$$\text{由勾股定理,得 } AB = \sqrt{AD^2 + BD^2} = \sqrt{12^2 + 5^2} = 13,$$

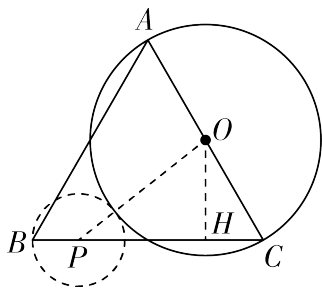
$\therefore$ 顶点 C 在 $\odot B$ 内, 顶点 A 在 $\odot B$ 外,

$$\therefore 10 < r < 13.$$

故答案为: $10 < r < 13$.

4. 【答案】 $\frac{4}{5}$

【解析】如图,



连接 OP , 过点 O 作 $OH \perp BC$ 于 P ,

在等边 $\triangle ABC$ 中, $AB = 4$,

$$\therefore AC = BC = AB = 4, \angle ACB = 60^\circ,$$

$\therefore$ 点 O 是 AC 的中点,

$$\therefore AO = OC = 2,$$

$\therefore$ 以线段 PB 为半径的 $\odot P$ 与以边 AC 为直径的 $\odot O$ 外切,

$$\therefore PO = 2 + BP,$$

$$\therefore OH \perp BC,$$

$$\therefore \angle COH = 30^\circ,$$

$$\therefore HC = 1, OH = \sqrt{3},$$

$$\therefore OP^2 = OH^2 + PH^2,$$

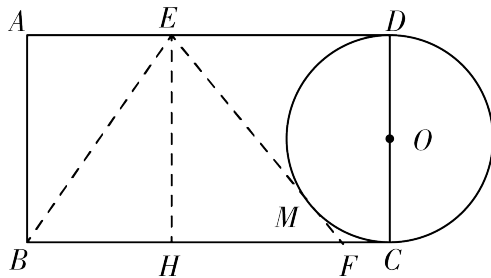
$$\therefore (2 + BP)^2 = 3 + (4 - 1 - BP)^2$$

$$\therefore BP = \frac{4}{5}.$$

故答案为: $\frac{4}{5}$.

5. 【答案】 3

【解析】如图, 设 $\odot O$ 与 EF 相切于 M , 连接 EB , 作 $EH \perp BC$ 于 H .



由题意易知四边形 $AEHB$ 是矩形, 设 $AE = BH = x$,

由切线长定理可知, $ED = EM, FC = FM$,

$\therefore B, F$ 关于 EH 对称,

$$\therefore HF = BH = x, ED = EM = 7 - x, FC =$$

$$FM = 7 - 2x, EF = 14 - 3x,$$

在 $\text{Rt}\triangle EFH$ 中, $\therefore EF^2 = EH^2 + HF^2$,

$$\therefore 4^2 + x^2 = (14 - 3x)^2,$$

$$\text{解得 } x = 3 \text{ 或 } \frac{15}{2} (\text{舍去}),$$

$$\therefore AE = 3,$$

故答案为 3.

6. 【答案】① $\frac{15}{8}$.

② 两圆内切, 证明见解析.

【解析】① 设圆 P 与直线 BC 相切于 E ,

连接 PE ,

$$\therefore PE \perp BC,$$

$$\because AB \perp BC, DC \perp BC,$$

$$\therefore AB \parallel CD,$$

$$\therefore \triangle ABP \sim \triangle CDP,$$

$$\therefore \frac{AB}{CD} = \frac{AP}{CP} = \frac{3}{5},$$

$$\therefore \frac{CP}{AC} = \frac{5}{8},$$

又 $\because PE \perp BC$,

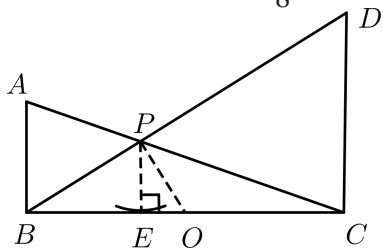
$$\therefore \angle PEC = \angle ABC = 90^\circ,$$

$$\therefore \triangle ABC \sim \triangle PEC,$$

$$\therefore \frac{PE}{AB} = \frac{CP}{CA} = \frac{5}{8},$$

$$\therefore PE = \frac{5}{8}AB = \frac{15}{8},$$

即圆 P 的半径长为 $PE = \frac{15}{8}$.



② 由题意得, 以 BC 为直径作圆 O , 则 O 为 BC 中点, 连接 OP ,

$$\text{在 Rt}\triangle BCD \text{ 中, } BD = \sqrt{BC^2 + CD^2} = \sqrt{64 + 25} = \sqrt{89},$$

$$\text{由①得 } \frac{BP}{PD} = \frac{AB}{CD} = \frac{3}{5},$$

$$\therefore \frac{PB}{BD} = \frac{3}{8},$$

$$\therefore PB = \frac{3}{8}BD = \frac{3\sqrt{89}}{8},$$

$$\text{设 } OE = x, BO = \frac{1}{2}BC = 4, BE = 4 - x,$$

$$CE = 4 + x,$$

$$\text{同理 } \frac{BE}{EC} = \frac{BP}{PD} = \frac{3}{5},$$

$$\text{即 } \frac{4-x}{4+x} = \frac{3}{5}, \text{ 解得 } x = 1,$$

$$\therefore OE = 1,$$

$$\text{在 Rt}\triangle PEO \text{ 中, } PO = \sqrt{PE^2 + EO^2} = \frac{17}{8},$$

$$\text{已知 } r_{\odot P} = PE = \frac{15}{8}, r_{\odot O} = OB = 4,$$

$$r_{\odot O} - r_{\odot P} = 4 - \frac{15}{8} = \frac{17}{8} = PO,$$

两圆半径之差等于圆心距, 两圆内切.

7. 【答案】 $2\sqrt{5} - 2$

【解析】 $\because$ 正多边形的对称轴共有 10 条,

$\therefore$ 这个正多边形是正十边形,

设这个正十边形的中心为 O , 则

$$OA = OB = 4,$$

$$\angle AOB = \frac{360^\circ}{10} = 36^\circ,$$

$$\because OA = OB,$$

$$\therefore \angle OAB = \angle B = 72^\circ,$$

作 AC 平分 $\angle OAB$ 交 OB 于 C ,

则 $\angle OAC = \angle O, \angle ACB = \angle B$,

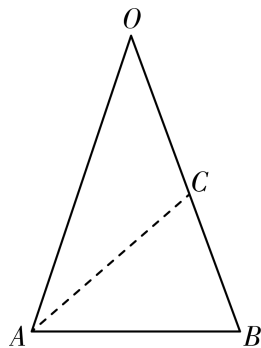
$$\therefore OC = CA = AB, \triangle ABC \sim \triangle OAB,$$

$$\therefore \frac{AB}{OA} = \frac{BC}{AB}, \text{ 即 } AB^2 = 4 \times (4 - AB),$$

$$\text{解得, } AB_1 = 2\sqrt{5} - 2, AB_2 = -2\sqrt{5} - 2 (\text{舍去}),$$

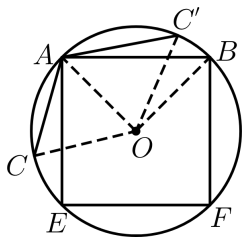
$$\therefore AB = 2\sqrt{5} - 2,$$

故答案为: $2\sqrt{5} - 2$.



8. 【答案】 15° 或 105°

【解析】 $\because$ 圆内接正六边形的中心角 $\angle AOC = 60^\circ$, 圆内接正四边形的中心角 $\angle AOB = 90^\circ$,



$$AO = OB,$$

$$AO = OC,$$

$$\therefore \angle CAO = 60^\circ,$$

$$\angle OAB = \angle OBA = 45^\circ,$$

$$\therefore \angle BAC = \angle CAO + \angle OAB = 60^\circ + 45^\circ = 105^\circ,$$

$$\text{或 } \angle BAC' = \angle C'AO - \angle OAB = 60^\circ - 45^\circ = 15^\circ,$$

故答案为: 15° 或 105° .

第5讲 代数综合(一)

1. 【答案】D

【解析】 $\because xy > 0$,

$\therefore x$ 和 y 同号,

$$\because x\sqrt{\frac{-y}{x^2}} \text{ 中, } \frac{-y}{x^2} \geq 0,$$

$$\therefore y < 0,$$

$$\therefore x < 0,$$

$$\therefore x\sqrt{\frac{-y}{x^2}} = -\sqrt{x^2 \cdot \frac{-y}{x^2}} = -\sqrt{-y}.$$

故选 D.

2. 【答案】 $-x^2 + x$, $x = 0$ 时, 原式 $= 0$

$$\begin{aligned} \text{【解析】原式} &= \left(\frac{2x-1}{x-1} - \frac{x^2-1}{x-1} \right) \div \frac{x-2}{(x-1)^2} \\ &= \frac{2x-1-x^2+1}{x-1} \cdot \frac{(x-1)^2}{x-2} \end{aligned}$$

$$\begin{aligned} &= \frac{x(2-x)}{x-1} \cdot \frac{(x-1)^2}{x-2} \\ &= -x \cdot (x-1) = -x^2 + x, \end{aligned}$$

$$\because x-1 \neq 0, x-2 \neq 0,$$

$$\therefore x \neq 1, x \neq 2,$$

$$\because 0 \leq x \leq 2,$$

$$\therefore x = 0,$$

$$\text{原式} = -0 + 0 = 0.$$

3. 【答案】4

【解析】方程 $(x^2 - 1) = 0$,

解得 $x = 1$ 或 $x = -1$,

因为方程 $(x^2 - 1)(kx^2 - 6x - 8) = 0$ 有三个不同实数解,

所以方程 $kx^2 - 6x - 8 = 0$ 的根需满足以下情况之一:

1. 有一个不等于 1 或 -1 的实数根

($k = 0$);

2. 有两个相等的实数根, 且该根不等于 1 或 -1 ($k \neq 0, \Delta = 0$);

3. 有两个不等的实数根, 其中一根为 1 或 -1, 另一根不为 1 或 -1.

当 $k = 0$ 时, 方程 $kx^2 - 6x - 8 = 0$ 变为 $-6x - 8 = 0$,

$$\text{解得 } x = -\frac{4}{3},$$

此时方程有三个不同的实数解 $x = 1$, $x = -1$, $x = -\frac{4}{3}$;

当 $k \neq 0$ 时, 方程 $kx^2 - 6x - 8 = 0$ 是一元二次方程,

若方程有两个相等的实数根, 则 $\Delta =$

$$(-6)^2 - 4k \times (-8) = 36 + 32k = 0,$$

解得 $k = -\frac{9}{8}$,

此时方程的根为 $x = \frac{6}{2k} = -\frac{8}{3}$, 与 $x = 1$, $x = -1$ 不同, 方程有三个不同的实数解;

若方程有一个根为 $x = 1$, 代入

$$kx^2 - 6x - 8 = 0 \text{ 得 } k - 6 - 8 = 0,$$

解得 $k = 14$, 此时方程为 $14x^2 - 6x - 8 = 0$,

$$\text{即 } 7x^2 - 3x - 4 = 0,$$

因式分解为 $(7x+4)(x-1) = 0$, 解得 $x = 1$

$$\text{或 } x = -\frac{4}{7},$$

此时方程有三个不同的实数解;

若方程有一个根为 $x = -1$, 代入 $kx^2 - 6x - 8 = 0$ 得 $k + 6 - 8 = 0$,

解得 $k = 2$, 此时方程为 $2x^2 - 6x - 8 = 0$, 即 $x^2 - 3x - 4 = 0$,

因式分解为 $(x-4)(x+1) = 0$, 解得 $x = -1$

$$\text{或 } x = 4,$$

此时方程有三个不同的实数解.

综上, k 的值有 4 个.

故答案为: 4.

4. 【答案】-1

【解析】 $\because x_1, x_2$ 是关于 x 的方程

$$x^2 + (2a-1)x + a^2 = 0 \text{ 的两个实数根,}$$

$$\therefore x_1 + x_2 = 1 - 2a, x_1 x_2 = a^2,$$

$$\therefore (x_1 + 2)(x_2 + 2) = 11, \text{ 即}$$

$$x_1 x_2 + 2(x_1 + x_2) + 4 = 11,$$

$$\therefore a^2 + 2(1 - 2a) + 4 = 11, \text{ 整理得:}$$

$$a^2 - 4a - 5 = 0,$$

解得: $a = -1$ 或 5 ,

当 $a = -1$ 时, 原方程化为 $x^2 - 3x + 1 = 0$,

$\Delta = 9 - 4 = 5 > 0, a = -1$ 符合题意;

当 $a = 5$ 时, 原方程化为 $x^2 + 9x + 25 = 0$,

$$\Delta = 81 - 100 = -19 < 0, a = 5 \text{ 不符合题意.}$$

故答案为: -1.

5. 【答案】A

$$\text{【解析】由 } \begin{cases} x > 2a - 3 \\ 2x \geq 3(x - 2) + 5 \end{cases},$$

解得: $2a - 3 < x \leq 1$,

又因为关于 x 的不等式组

$$\begin{cases} x > 2a - 3 \\ 2x \geq 3(x - 2) + 5 \end{cases} \text{ 仅有三个整数解,}$$

故解得: $-2 \leq 2a - 3 < -1$,

$$\text{解得: } \frac{1}{2} \leq a < 1,$$

故选: A.

6. 【答案】B

【解析】 $\because$ 一次函数 $y = kx - 3k + 4$,

$\therefore k \neq 0$, 故 ① 正确;

当 $k < 0$ 时, $-3k + 4 > 4 > 0$,

$\therefore$ 一次函数 $y = kx - 3k + 4$ 的图象经过第一、二、四象限, 故 ② 正确;

当 $k = 2$ 时, 一次函数 $y = kx - 3k + 4$ 化为 $y = 2x - 2$,

当 $x = 0$ 时, $y = -2$, 当 $y = 0$ 时, $x = 1$,

$\therefore y = 2x - 2$ 与 y 轴交点为 $(0, -2)$, 与 x 轴交点为 $(1, 0)$, 则图象与坐标轴围成三角形的面积为 $\frac{1}{2} \times 2 \times 1 = 1$, 故 ③ 错误;

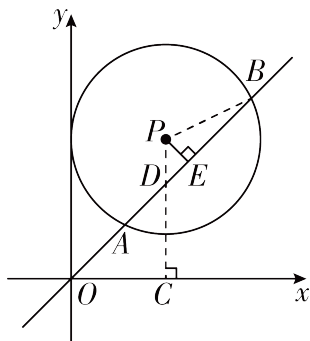
当 $x = 3$ 时, $y = kx - 3k + 4 = 3k - 3k + 4 = 4$,

$\therefore$ 不论 k 为何值, 其图象一定经过一个定点 $(3, 4)$, 故 ④ 正确.

故选 B.

7. 【答案】 $3 + \sqrt{2}$

【解析】作 $PC \perp x$ 轴于 C , 交 AB 于 D , 作 $PE \perp AB$ 于 E , 连接 PB , 如图,



$\because \odot P$ 的圆心坐标是 $(3, a)$,

$\therefore OC = 3, PC = a$, 把 $x = 3$ 代入 $y = x$ 得 $y = 3$,

$\therefore D$ 点坐标为 $(3, 3)$,

$\therefore CD = 3$,

$\therefore \triangle OCD$ 为等腰直角三角形,

$\therefore \triangle PED$ 也为等腰直角三角形,

$\because PE \perp AB$,

$\therefore AE = BE = \frac{1}{2}AB = \frac{1}{2} \times 4\sqrt{2} = 2\sqrt{2}$,

在 $\text{Rt}\triangle PBE$ 中, $PB = 3$,

$\therefore PE = \sqrt{3^2 - (2\sqrt{2})^2} = 1$,

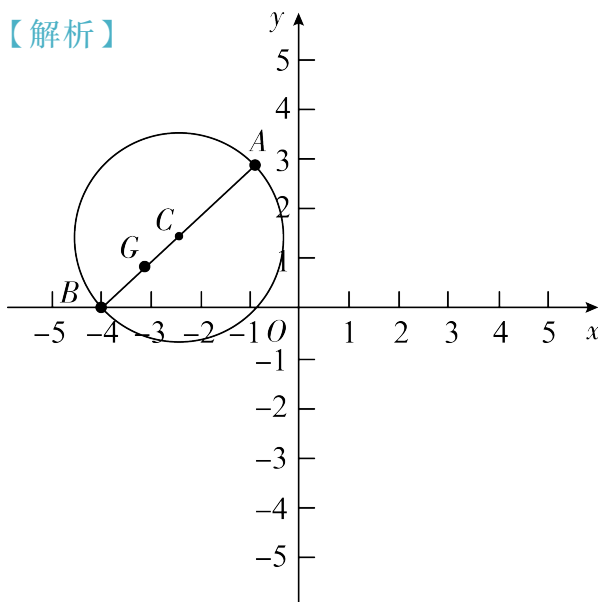
$\therefore PD = \sqrt{2}PE = \sqrt{2}$,

$\therefore a = 3 + \sqrt{2}$.

故答案为: $3 + \sqrt{2}$.

8. 【答案】 $\sqrt{2}$

【解析】



设直线 AB 的解析式为 $y = kx + b$,

把 $A(-1, 3), B(-4, 0)$ 代入得

$$\begin{cases} -k + b = 3 \\ -4k + b = 0 \end{cases},$$

$$\text{解得} \begin{cases} k = 1 \\ b = 4 \end{cases},$$

$\therefore$ 直线 $AB: y = x + 4$,

把 $x = -3$ 代入得 $y = 1$,

$\therefore$ 点 G 在直径 AB 上,

$\therefore$ 点 G 与 $\odot C$ 的最短距离为 GB ,

$\therefore GB = \sqrt{(-3+4)^2 + 1^2} = \sqrt{2}$,

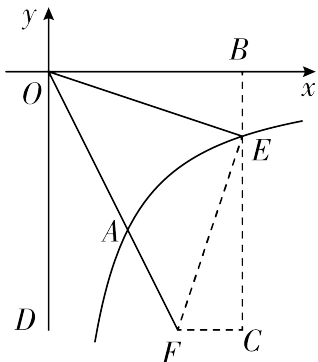
$\therefore$ 点 G 与 $\odot C$ 的最短距离为 $\sqrt{2}$.

故答案为: $\sqrt{2}$.

第6讲 代数综合(二)

1. 【答案】 $y = -2x$

【解析】如图所示,过 E 作 $EF \perp EO$ 交 OA 于点 F ,



$$\because \angle AOE = 45^\circ, EO \perp EF,$$

$$\therefore \angle OEF = 90^\circ, OE = FE,$$

过点 E 作 y 轴的平行线交 x 轴于点 B , 作 $FC \perp BE$ 于 C ,

$$\therefore \angle OBE = \angle ECF = 90^\circ,$$

$$\text{又 } \because \angle BEO + \angle BOE = 90^\circ,$$

$$\angle BEO + \angle CEF = 90^\circ,$$

$$\therefore \angle BOE = \angle CEF,$$

在 $\triangle BOE$ 和 $\triangle CEF$ 中,

$$\begin{cases} \angle OBE = \angle ECF \\ \angle BOE = \angle CEF \\ OE = EF \end{cases}$$

$$\therefore \triangle BOE \cong \triangle CEF (\text{AAS}),$$

$$\therefore OB = EC, BE = CF.$$

$$\text{又 } \because E(6, -2),$$

$$\therefore OB = EC = 6, BE = CF = 2, \text{ 即:}$$

$$BC = 2 + 6 = 8,$$

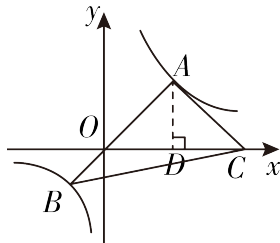
$$\therefore \text{点 } F \text{ 的坐标为 } (4, -8),$$

$\therefore$ 设直线 OA 的表达式为: $y = ax$, 代入 $F(4, -8)$ 可得 $4a = -8, a = -2$,

$\therefore$ 直线 OA 的表达式为: $y = -2x$.

2. 【答案】 C

【解析】如图,过 A 作 $AD \perp x$ 轴于 D .



由题意, 设 $A\left(a, \frac{k}{a}\right) (a > 0)$,

$$\because AO = AC, AD \perp OC,$$

$$\therefore OC = 2OD = 2a,$$

又设直线 OA 的解析式为 $y = mx$,

$$\therefore ma = \frac{k}{a},$$

$$\therefore m = \frac{k}{a^2},$$

$$\therefore \text{直线 } OA \text{ 的解析式为 } y = \frac{k}{a^2}x,$$

$$\text{联立 } \begin{cases} y = \frac{k}{4x} \\ y = \frac{k}{a^2}x \end{cases},$$

$$\text{得 } x^2 = \frac{a^2}{4},$$

$$\therefore x = \pm \frac{a}{2},$$

$$\therefore B\left(-\frac{a}{2}, -\frac{k}{2a}\right),$$

$$\therefore S_{\triangle ABC} = S_{\triangle BOC} + S_{\triangle AOC}$$

$$= \frac{1}{2}OC \cdot |y_B| + \frac{1}{2}OC \cdot |y_A|$$

$$= \frac{1}{2} \times 2a \left(\frac{k}{2a} + \frac{k}{a} \right)$$

$$= \frac{3}{2}k,$$

又 $\because S_{\triangle ABC} = 6$,

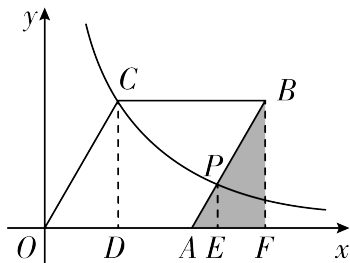
$$\therefore \frac{3}{2}k = 6,$$

$$\therefore k = 4.$$

故选: C.

3. 【答案】 $\sqrt{2} - 1$

【解析】作 $CD \perp x$ 轴, $BF \perp x$ 轴, $PE \perp x$ 轴, 垂足分别为 D, F, E ,



设菱形边长为 a , AP 的长为 b ,

$$\because \angle COA = 60^\circ,$$

$$\text{则 } C\left(\frac{a}{2}, \frac{\sqrt{3}}{2}a\right), P\left(a + \frac{b}{2}, \frac{\sqrt{3}}{2}b\right),$$

$$B\left(\frac{3}{2}a, \frac{\sqrt{3}}{2}a\right),$$

把 C, P 坐标代入反比例函数 $y = \frac{k}{x}$, 得

$$k = \frac{a}{2} \times \frac{\sqrt{3}}{2}a = \left(a + \frac{b}{2}\right) \left(\frac{\sqrt{3}}{2}b\right),$$

$$\therefore b = (\sqrt{2} - 1)a,$$

$$\therefore \frac{AP}{AB} = \frac{PE}{BF} = \frac{\frac{\sqrt{3}}{2}b}{\frac{\sqrt{3}}{2}a} = \frac{b}{a} = \sqrt{2} - 1.$$

4. 【答案】 B

【解析】设点 M 坐标为 $\left(a, \frac{1}{a}\right)$, 点 N 坐标为 $\left(b, \frac{1}{b}\right)$,

$$\text{则 } A(a, 0), B\left(0, \frac{1}{b}\right), C\left(a, \frac{1}{b}\right),$$

$$\therefore OB = AC = \frac{1}{b}, OA = BC = a, BN = b,$$

$$AM = \frac{1}{a}, CN = a - b, CM = \frac{1}{b} - \frac{1}{a},$$

$$\therefore S_{\triangle COM} = \frac{1}{2}CM \times OA = \frac{1}{2} \times \left(\frac{1}{b} - \frac{1}{a}\right) \times a = \frac{a}{2b} - \frac{1}{2},$$

$$S_{\triangle CON} = \frac{1}{2}CN \times OB = \frac{1}{2} \times (a - b) \times \frac{1}{b} = \frac{a}{2b} - \frac{1}{2},$$

$\therefore S_{\triangle COM} = S_{\triangle CON}$, 故结论 ① 正确;

$$S_{\triangle MCN} = \frac{1}{2}CN \cdot CM = \frac{1}{2} \times (a - b) \left(\frac{1}{b} - \frac{1}{a}\right) = \frac{(a - b)^2}{2ab},$$

$$S_{\triangle MON} = S_{\text{矩形}OACB} - S_{\triangle OBN} - S_{\triangle OAM} -$$

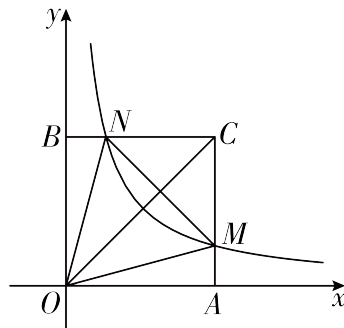
$$S_{\triangle MCN} \\ = a \times \frac{1}{b} - \frac{1}{2} \times b \times \frac{1}{b} - \frac{1}{2} \times a \times \frac{1}{a} - \frac{(a - b)^2}{2ab} \\ = \frac{a^2 - b^2}{2ab},$$

当 $\triangle MON$ 与 $\triangle MCN$ 的面积相等时, $\frac{(a - b)^2}{2ab} = \frac{a^2 - b^2}{2ab}$, 即 $a = b$,

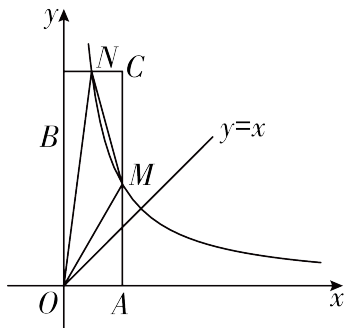
当 $a = b$ 时, M, N 重合, 与题意不符, 故结论 ② 错误;

$\because$ 等边三角形和反比例函数都是轴对称图形, 当 $\angle NOM = 60^\circ$ 且对称轴都为直线 $y = x$, $\triangle MON$ 可能是等边三角形, 故 ④ 正确,

如图:



当 M, N 在 $y=x$ 的同侧时, $\triangle MON$ 可能是钝角三角形, 故 ③ 错误;



综上, ①④ 正确、②③ 错误.

故选: B.

5. 【答案】D

【解析】A、当 $-1 < x < 3$ 时, 根据函数图象可得: $y > 0$,

故该选项错误.

B、 $\because$ 对称轴经过点 $(1, 0)$,

$$\therefore -\frac{b}{2a} = 1,$$

故该选项错误.

C、二次函数 $y = ax^2 + bx + 3$ 的图象与 x 轴交于 $A(-1, 0), B(3, 0)$ 两点,

将 $A(-1, 0), B(3, 0)$ 代入 $y = ax^2 + bx + 3$ 中,

$$\text{可得: } \begin{cases} a - b + 3 = 0 \\ 9a + 3b + 3 = 0 \end{cases},$$

$$\text{解得: } \begin{cases} a = -1 \\ b = 2 \end{cases},$$

函数解析式为: $y = -x^2 + 2x + 3$.

$$\text{将 } \begin{cases} a = -1 \\ b = 2 \end{cases} \text{ 代入 } ax^2 + bx + 3 = 5,$$

$$\text{可得: } -x^2 + 2x + 3 = 5,$$

$$\text{即 } -x^2 + 2x - 2 = 0,$$

$$a = -1, b = 2, c = -2,$$

$$\Delta = b^2 - 4ac = 4 - 4 \times (-1) \times (-2) = -4 < 0,$$

故方程没有实数根,

该选项错误.

D、 $y = -x^2 + 2x + 3 = -(x-1)^2 + 4$, 故二次函数 $y = ax^2 + bx + 3$ 的顶点坐标为 $(1, 4)$, 该选项正确.

故选 D.

6. 【答案】B

【解析】① $\because$ 图象与 x 轴交于点 $A(-1, 0)$, 对称轴为直线 $x = 1$,

$\therefore$ 图象与 x 轴的另一个交点为 $(3, 0)$,

$\therefore$ 当 $x = 4$ 时, $y > 0$,

$\therefore 16a + 4b + c > 0$, 故 ① 错误;

② $\because$ 对称轴为直线 $x = -\frac{b}{2a} = 1$,

$$\therefore b = -2a,$$

$\because$ 图象与 y 轴的交点 B 在 $(0, -2)$ 和 $(0, -1)$ 之间,

$$\therefore -2 < c < -1,$$

$$\therefore 1 < -c < 2,$$

$$\therefore -c - 1 > 0,$$

又 $\because a > 0$,

$$\therefore b^2 - 4ac - 4a = 4a^2 - 4ac - 4a = 4a(a - c - 1) > 0,$$

$$\therefore b^2 - 4ac > 4a, \text{ 即 ② 正确.}$$

③ $\because$ 图象与 y 轴的交点 B 在 $(0, -2)$ 和 $(0, -1)$ 之间,

$$\therefore -2 < c < -1,$$

$\therefore$ 图象与 x 轴交于点 $A(-1, 0)$ 和 $(3, 0)$,

$\therefore ax^2 + bx + c = 0$ 的两根为 -1 和 3 ,

$$\therefore -3 = \frac{c}{a},$$

$$\therefore c = -3a,$$

$$\therefore -2 < -3a < -1,$$

$$\therefore \frac{1}{3} < a < \frac{2}{3}, \text{故 ③ 正确;}$$

$$\text{④} \because \text{对称轴为直线 } x = -\frac{b}{2a} = 1,$$

$$\therefore b = -2a,$$

$$\therefore a > 0, c = -3a,$$

$$\therefore b > c, \text{故 ④ 错误.}$$

综上所述, 正确的有 ②③,

故选 B.

7. (1) 【答案】 $(m, m+1)$

【解析】 $\because y = x^2 - 2mx + m^2 + m + 1$,

$$\therefore y = (x - m)^2 + m + 1,$$

$\therefore$ 图象 C 的顶点坐标为 $(m, m+1)$.

(2) 【答案】 $6 - \sqrt{19} < m \leq 6 + \sqrt{19}$ 或 $-2 \leq m \leq -1$ 或 $m = \frac{2\sqrt{5}}{5}$ 时, 图象 C 与线段 AB 恰有一个公共点

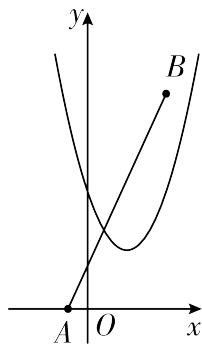
【解析】 ① 当 $m > 0$ 时, 设点 A 的坐标为 $(-1, 0)$, 点 B 的坐标为 $(4, 5m)$,

一次函数 $y = mx + m$ (m 是常数, $m \neq 0$, $-1 \leq x \leq 4$) 的图象为线段 AB,

当 $x = -1$ 时, $y = 0$, 当 $x = 4$ 时, $y = 5m$,

依题意, 图象 C 与线段 AB 恰有一个公共点,

如图,



$$\text{当 } x = -1 \text{ 时, } y = (-1 - m)^2 + m + 1 \geq 0,$$

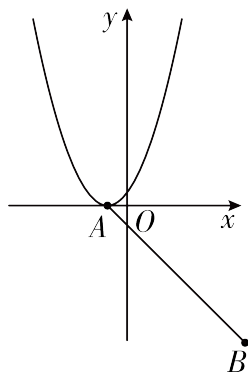
$$\text{解得: } m \geq -1 \text{ 或 } m \leq -2,$$

$$\text{当 } x = 4 \text{ 时, } y = (4 - m)^2 + m + 1 \leq 5m,$$

$$\text{解得: } 6 - \sqrt{19} < m \leq 6 + \sqrt{19},$$

$$\therefore 6 - \sqrt{19} < m \leq 6 + \sqrt{19};$$

② 当 $m < 0$ 时,



$$\begin{cases} (-1 - m)^2 + m + 1 \leq 0 \\ (4 - m)^2 + m + 1 \geq 5m \end{cases},$$

$$\text{解得: } -2 \leq m \leq -1,$$

当一次函数 $y = mx + m$ 与二次函数 $y = x^2 - 2mx + m^2 + m + 1$ 联立方程, 得 $x^2 - 2mx + m^2 + m + 1 = mx + m$, 一元二次方程有且只有两个相等实数根时:

$$\text{整理得 } x^2 - 3mx + m^2 + 1 = 0,$$

$$\Delta = (3m)^2 - 4 \times (m^2 + 1) = 0,$$

$$\text{解得 } m = \pm \frac{2\sqrt{5}}{5},$$

当 $m = -\frac{2\sqrt{5}}{5}$ 时, 交点横坐标为 $x = -\frac{3\sqrt{5}}{5}$
(不在 x 取值范围舍去),

综上所述, $6 - \sqrt{19} < m \leq 6 + \sqrt{19}$ 或 $-2 \leq m \leq -1$ 或 $m = \frac{2\sqrt{5}}{5}$ 时, 图象 C 与线段 AB 恰有一个公共点.

第7讲 自招专题训练(一)

1. 【答案】9

【解析】 $n^{12} - n^8 - n^4 + 1$
 $= n^8(n^4 - 1) - (n^4 - 1)$
 $= (n^4 - 1)(n^8 - 1)$
 $= (n^2 + 1)(n^2 - 1)(n^4 + 1)(n^4 - 1)$
 $= (n^4 + 1)(n^2 + 1)^2(n + 1)^2(n - 1)^2,$
 $\because n$ 是正奇数
 $\therefore (n^4 + 1), (n^2 + 1)$ 为 2 的倍数,
 $(n + 1), (n - 1)$ 都为 2 的倍数至少有一个
 为 4 的倍数,
 $\therefore$ 至少含有 $1 + 2 + 2 + 4 = 9$ 个因数 2
 经验证, 当 $n = 3$ 时,
 原式 $= 82 \times 10^2 \times 4^2 \times 2^2$
 $= 41 \times 5^2 \times 2^9$
 $\therefore k$ 的最大值为 9.

2. 【答案】4

【解析】将原方程变形为: $x^{284} + y^2 - 2y + 1 - 1 = 0$,
 即 $x^{284} + (y - 1)^2 = 1$,
 $\because x^{284} \geq 0, (y - 1)^2 \geq 0$, 且 x, y 是整数,
 $\therefore$ ① 当 $x = 1$ 时, $y = 1$;
 ② 当 $x = -1$ 时, $y = 1$;
 ③ 当 $x = 0$ 时, $y = 2$;

④ 当 $x = 0$ 时, $y = 0$,

共 4 组解.

故答案为: 4.

$$3. 【答案】 \begin{cases} x_1 = 2 \\ y_1 = 3 \\ z_1 = 4 \end{cases}, \begin{cases} x_2 = -2 \\ y_2 = -3 \\ z_2 = -4 \end{cases}$$

【解析】由原方程组变为

$$\begin{cases} (x + y - z)(x - y + z) = 3 \\ (y + z - x)(y - z + x) = 5 \\ (z + x - y)(z - x + y) = 15 \end{cases},$$

设 $x + y - z = a, x - y + z = b, y + z - x = c$,

$$\therefore \begin{cases} ab = 3 \\ ac = 5 \\ bc = 15 \end{cases},$$

$$\text{解得: } \begin{cases} a = 1 \\ b = 3 \\ c = 5 \end{cases} \text{ 或 } \begin{cases} a = -1 \\ b = -3 \\ c = -5 \end{cases},$$

$$\therefore \begin{cases} x + y - z = 1 \\ x - y + z = 3 \\ y + z - x = 5 \end{cases} \text{ 或 } \begin{cases} x + y - z = -1 \\ x - y + z = -3 \\ y + z - x = -5 \end{cases},$$

$$\text{解得: } \begin{cases} x_1 = 2 \\ y_1 = 3 \\ z_1 = 4 \end{cases}, \begin{cases} x_2 = -2 \\ y_2 = -3 \\ z_2 = -4 \end{cases}.$$

4. 【答案】C

【解析】 $\because x^2 + p_1x + q_1 = 0$ 与

$$x^2 + p_2x + q_2 = 0,$$

$$\therefore \Delta_1 = p_1^2 - 4q_1, \Delta_2 = p_2^2 - 4q_2,$$

两式相加得:

$$\Delta_1 + \Delta_2 = p_1^2 + p_2^2 - 4q_1 - 4q_2 =$$

$$p_1^2 + p_2^2 - 4(q_1 + q_2),$$

$$\text{又 } \because p_1 p_2 = 2(q_1 + q_2),$$

$$\therefore \Delta_1 + \Delta_2 = (p_1 - p_2)^2 \geq 0,$$

所以 Δ_1, Δ_2 中至少有一个大于等于 0.

故答案为 C.

5. 【答案】 $2\sqrt{5}$

【解析】原式 $= (a+b+c)^2 - b^2 - c^2 + 2bc = 20$

$$(a+b+c)^2 = (b-c)^2 + 20 \geq 20$$

$$\because a+b+c > 0$$

$$\therefore (a+b+c)_{\min} = 2\sqrt{5}.$$

6. 【答案】 $-3 \leq t \leq -\frac{1}{3}$

【解析】由 $a^2 + ab + b^2 = 1, ab - a^2 - b^2 = t$ 相加, 得 $2ab = t+1$,

$$\text{故 } ab = \frac{t+1}{2},$$

$$\text{又 } (a+b)^2 = a^2 + ab + b^2 + ab = 1 + ab = \frac{t+3}{2} \geq 0,$$

于是可知 a, b 是关于 x 的方程 $x^2 \pm \sqrt{\frac{t+3}{2}}x + \frac{t+1}{2} = 0$ 的两个实数根,

$$\text{所以 } \Delta = \frac{t+3}{2} - 2(t+1) = -\frac{3}{2}t - \frac{1}{2} \geq 0,$$

$$\text{解出 } t \leq -\frac{1}{3},$$

综上所述, t 的取值范围是 $-3 \leq t \leq -\frac{1}{3}$.

7. 【答案】 $\frac{8}{81}$

【解析】因为 $\frac{7}{9} = xy + x + y \geq xy + 2\sqrt{xy}$, 所以 $(\sqrt{xy} + 1)^2 \leq \frac{16}{9}, 0 < xy \leq \frac{1}{9}$.

$$\text{所以 } -x^2y^2 + xy = -\left(xy - \frac{1}{2}\right)^2 + \frac{1}{4} \leq -$$

$$\left(\frac{1}{9} - \frac{1}{2}\right)^2 + \frac{1}{4} = \frac{8}{81},$$

所以, 当 $xy = \frac{1}{9}$ 时, 即 $x = y = \frac{1}{3}$ 时, $-x^2y^2 + xy$ 取最大值为 $\frac{8}{81}$.

8. 【答案】 A

【解析】 $\because$ 四个小朋友去取球, 每个小朋友至少取一个球, 所有的球都被取完, 即将 10 个球分成了 4 份:



10 个球有 9 个空隙, 选 3 个空隙插上“隔板”即可分成 4 份,

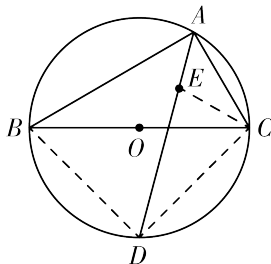
$$\text{即 } C_9^3 = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84 \text{ 种,}$$

故选: A.

第 8 讲 自招专题训练(二)

1. 【答案】 $2\sqrt{2}$

【解析】如图, 连接 BD, CD, EC .



$\because$ 点 E 是 $\triangle ABC$ 的内心,

$$\therefore \angle DAB = \angle DAC, \angle ECA = \angle ECB,$$

$$\because \angle DCB = \angle DAB, \angle DEC = \angle EAC + \angle ECA, \angle ECD = \angle ECB + \angle DCB,$$

$$\because \angle BCD = 162^\circ,$$

$$\therefore \angle 1 = 180^\circ - 162^\circ = 18^\circ.$$

$$\text{而 } \angle ABC = 78^\circ,$$

$$\therefore \angle 2 = 78^\circ - 18^\circ = 60^\circ,$$

$$\text{又 } \because AB = BC,$$

$$\therefore BA = BG,$$

$$\therefore \triangle ABG \text{ 为等边三角形},$$

$$\therefore GA = GB,$$

$$\therefore GD = GA,$$

$$\text{又 } \because \angle BGD = \angle BCD = 162^\circ,$$

$$\text{而 } \angle AGB = 60^\circ,$$

$$\therefore \angle AGD = 360^\circ - 162^\circ - 60^\circ = 138^\circ,$$

$$\therefore \angle 4 = \angle 5 = \frac{1}{2}(180^\circ - 138^\circ) = 21^\circ,$$

$$\text{而 } GD \parallel BC,$$

$$\therefore \angle AEB = \angle 4 = 21^\circ.$$

故答案为 21° .

5. 【答案】见解析

【解析】在 AC 上截 $AM = AB$, 连接 DM ,

易证 $\triangle ABD \cong \triangle AMD$,

于是 $\angle AMD = \angle B = 2\angle C$, 即 $DM = MC$,

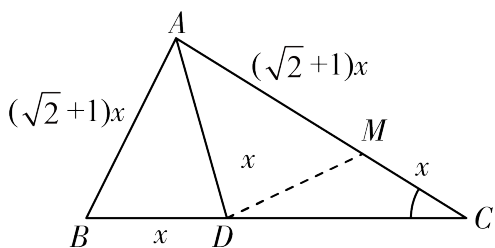
$$\text{又 } \frac{AB}{BD} = \sqrt{2} + 1 = \frac{AC}{CD},$$

设 $BD = DM = MC = x$, 则 $AB = AM =$

$$(\sqrt{2} + 1)x,$$

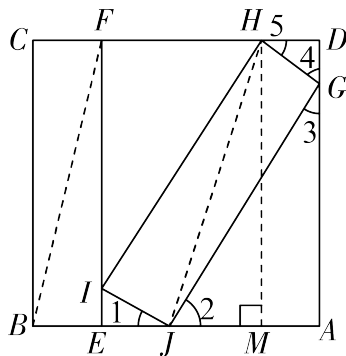
$$\text{于是 } DC = \frac{(\sqrt{2} + 2)x}{\sqrt{2} + 1} = \sqrt{2}x,$$

$$\tan C = 1.$$



6. 【答案】 $2 - \sqrt{3}$

【解析】过 H 作 $HM \perp AB$ 于 M , 分别连接 BF, JH ,



$$\because AB \parallel CD,$$

$$\therefore FE = HM.$$

$$\because \text{矩形 } BEFC \text{ 与 } IJGH \text{ 全等},$$

$$\therefore BF = JH.$$

$$\therefore \triangle BEF \cong \triangle JMH (\text{HL}).$$

$$\therefore BE = JM = IJ = HG.$$

$$\therefore \angle 1 + \angle 2 = \angle 2 + \angle 3 = \angle 3 + \angle 4 = \angle 4 + \angle 5.$$

$$\therefore \angle 1 = \angle 3 = \angle 5.$$

$$\because \angle FEJ = \angle D = 90^\circ, IJ = HG,$$

$$\therefore \triangle IEJ \cong \triangle GDH (\text{AAS}).$$

$$\therefore EJ = HD = MA.$$

$$\therefore BE + EJ = JM + MA = EJ + JM = \frac{1}{2}AB = \frac{1}{2}.$$

$$\text{设 } IJ = JM = x,$$

$$\triangle IEJ \sim \triangle JAG,$$

$$\therefore \frac{IE}{JA} = \frac{IJ}{JG}.$$

把 $IE = \frac{1}{2}x$, $EJ = \frac{1}{2} - x$ 代入 $IE^2 + EJ^2 = IJ^2$,

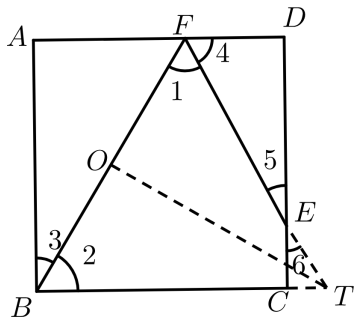
$$\therefore x = 2 + \sqrt{3} \text{ (舍) 或 } x = 2 - \sqrt{3},$$

即: $BE = 2 - \sqrt{3}$.

答案: $2 - \sqrt{3}$.

7. 【答案】 $\frac{3}{5}$.

【解析】延长 FE 交 BC 的延长线于 T , 设 FB 的中点为 O , 连 TO ,



$$\therefore \angle 1 = \angle 2,$$

$$\therefore FT = BT,$$

又 O 为 FB 中点,

$$\therefore OT \perp BF,$$

$\therefore$ 四边形 $ABCD$ 为正方形,

设 $AD = DC = BC = AB = 4k$,

$$\angle D = \angle ABC = \angle A = 90^\circ,$$

$$\therefore \angle 2 + \angle 3 = 90^\circ, \angle 2 + \angle OTB = 90^\circ,$$

$$\therefore \angle 3 = \angle OTB,$$

$$\text{又 } \angle A = \angle TOB = 90^\circ,$$

$$\therefore \triangle BAF \sim \triangle TOB,$$

$$\therefore \frac{AF}{OB} = \frac{BF}{BT},$$

$$\text{又 } OB = \frac{1}{2}BF,$$

$$\therefore \frac{AF}{\frac{1}{2}BF} = \frac{BF}{BT}, BF^2 = 2AF \cdot BT,$$

设 $CT = x (x > 0)$, $CD = AD = 4k$, $DE = 3EC$,

$$\therefore DE = 3k, EC = k,$$

又 $DF \parallel CT$,

$$\therefore \angle 4 = \angle ETC,$$

又 $\angle 5 = \angle 6$,

$$\therefore \triangle FDE \sim \triangle TCE,$$

$$\therefore \frac{DF}{CT} = \frac{DE}{EC} = 3,$$

$$\therefore DF = 3x, AF = 4k - 3x, BT = 4k + x,$$

$$\text{在 Rt}\triangle ABF \text{ 中, } BF^2 = AB^2 + AF^2 = (4k)^2 + (4k - 3x)^2,$$

$$\therefore \text{有 } (4k)^2 + (4k - 3x)^2 = 2(4k - 3x)(4k + x),$$

$$\text{整理有 } 15x^2 = 8kx,$$

又 $x > 0$,

$$\therefore 15x = 8k,$$

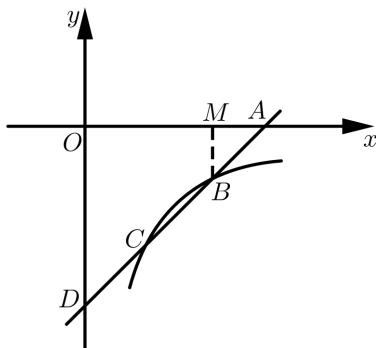
$$x = \frac{8}{15}k,$$

$$\therefore AF = 4k - \frac{8}{5}k = \frac{12}{5}k,$$

$$\therefore \tan \angle ABF = \frac{AF}{AB} = \frac{\frac{12}{5}k}{4k} = \frac{3}{5}.$$

8. 【答案】 $-\frac{9}{2}$

【解析】如图, 作 $BM \parallel y$ 轴,



$\because AB = BC = CD,$
 $\therefore BM = \frac{1}{3}OD, OM = \frac{2}{3}OA,$
 $\because$ 直线 $y = x + k (k < 0),$
 $\therefore A(-k, 0), D(0, k),$
 $\therefore OA = OD = -k,$
 $\therefore B\left(-\frac{2}{3}k, \frac{1}{3}k\right),$
 $\because B$ 点在双曲线 $y = \frac{k}{x}$ 上,
 $\therefore -\frac{2}{3}k \cdot \frac{1}{3}k = k,$
 解得 $k = -\frac{9}{2}.$
 故答案为: $-\frac{9}{2}.$

第9讲 三角形与四边形综合

1. (1)【答案】证明见解析

【解析】 $\because AD \parallel BC, \angle ABC = 90^\circ,$
 $\therefore \angle ADB = \angle CBE, \angle BAD = 90^\circ,$
 $\because CF \perp BD,$
 $\therefore \angle CEB = \angle BEF = 90^\circ,$
 在 $\triangle ABD$ 和 $\triangle ECB$ 中,
 $\because \angle CEB = \angle BAD = 90^\circ, \angle ADB = \angle EBC,$
 $BD = BC,$

$\therefore \triangle ABD \cong \triangle ECB.$

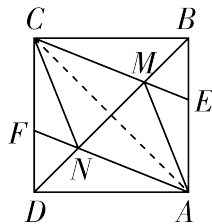
(2)【答案】 $\frac{\sqrt{2}-1}{2}$

【解析】 在 $\triangle ABD$ 和 $\triangle EBF$ 中,
 $\because \angle FBE = \angle ABD, \angle BAD = \angle BEF = 90^\circ,$
 $\therefore \triangle ABD \sim \triangle EBF,$
 $\therefore \frac{EF}{AD} = \frac{BE}{AB},$
 $\because \triangle ABD \cong \triangle ECB,$

$\therefore BE = AD = 4, AB = EC,$
 $\therefore EF \cdot AB = AD \cdot BE = 4 \times 4 = 16,$
 $\therefore EF = \frac{16}{AB},$
 又 $\because F$ 是 AB 的中点,
 $\therefore BF = \frac{1}{2}AB,$
 $\therefore EF = \frac{8}{BF},$
 在 $\text{Rt}\triangle BEF$ 中, $EF^2 + BE^2 = \frac{64}{BF^2} + 4^2 = BF^2,$
 $\therefore BF^2 = 8 + 8\sqrt{2}$ (舍负),
 $\therefore FE : EC = \frac{16}{AB \cdot EC} = \frac{16}{AB^2} = \frac{4}{BF^2} =$
 $\frac{4}{8 + 8\sqrt{2}} = \frac{\sqrt{2}-1}{2}.$

2. (1)【答案】证明见解析

【解析】 如图, 连接 $AC,$

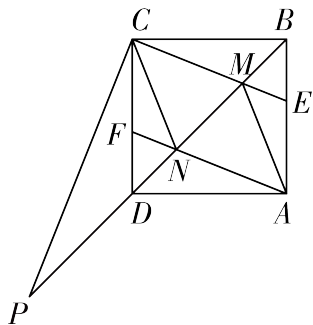


$\because$ 四边形 $ABCD$ 是正方形,
 $\therefore AB \parallel CD, AD \parallel BC, AB = CD = BC = AD,$
 $AC \perp BD, \angle ADC = \angle ABC = 90^\circ,$
 $\therefore AF = CE,$
 $\therefore \text{Rt}\triangle BEC \cong \text{Rt}\triangle DFA \text{ (HL)},$
 $\therefore BE = FD, \angle BCM = \angle DAN,$
 $\therefore AE = CF,$
 $\therefore$ 四边形 $AECF$ 是平行四边形,

$\therefore AF \parallel CE$,
 $\therefore AD \parallel BC$,
 $\therefore \angle ADB = \angle CBD$,
 $\therefore \triangle CBM \cong \triangle ADN \text{ (ASA)}$,
 $\therefore CM = AN$,
 四边形 $AMCN$ 是平行四边形,
 又 $\because AC \perp BD$, 即 $AC \perp MN$,
 $\therefore$ 平行四边形 $AMCN$ 是菱形.

(2) 【答案】证明见解析

【解析】如图,



$\because$ 四边形 $ABCD$ 是正方形, BD 是对角线,
 $\therefore \angle CDB = \angle CBD = 45^\circ$, $\angle BCD = 90^\circ$,
 由(1)得四边形 $AMCN$ 是菱形,
 $\therefore CM = CN$, 则 $\angle CNM = \angle CMN$,
 $\therefore \angle CND = \angle CMB$,
 $\therefore \triangle CND \cong \triangle CMB$,
 $\therefore \angle DCN = \angle BCM$,
 $\therefore CM^2 = MN \cdot NB$,
 $\therefore CN^2 = MN \cdot NB$
 $\therefore \frac{CN}{MN} = \frac{NB}{CN}$,

$\therefore \angle CNM = \angle BNC$,
 $\therefore \triangle CNM \sim \triangle BNC$,
 $\therefore \angle NCM = \angle NBC = 45^\circ$,
 $\therefore CP \perp CE$, 则 $\angle PCE = 90^\circ$,
 $\therefore \angle PCN = \angle CDN = 45^\circ$,
 $\therefore \angle CND = \angle PNC$,
 $\therefore \triangle NCD \sim \triangle NPC$,
 $\therefore \frac{CN}{PN} = \frac{CD}{PC}$, $\angle DCN = \angle P$,
 $\therefore \angle BCM + \angle DCM = \angle DCM + \angle DCP = 90^\circ$,
 $\angle DCN = \angle BCM$,
 $\therefore \angle BCM = \angle DCN = \angle DCP = \angle P$,
 $\therefore CD = DP$,
 $\therefore \frac{CN}{PN} = \frac{DP}{PC}$.

3. (1) 【答案】证明见解析

【解析】 $\because$ 四边形 $ABCD$ 是矩形,

$\therefore \angle A = \angle B = \angle C = \angle D = 90^\circ$, $AB = CD$,
 $AD = BC$,
 $\because EF \parallel AD$,
 $\therefore \angle AEF = \angle B = 90^\circ$, $\angle DFE = \angle C = 90^\circ$,
 $\angle BEF = \angle A = 90^\circ$, $\angle EFC = \angle D = 90^\circ$,
 $\therefore$ 四边形 $AEFD$ 和四边形 $EBCF$ 都是矩形,
 $\therefore AE = DF$, $AD = EF$, $BE = CF$, $EF = BC$,
 又 $\because EF^2 = AE \cdot BE$,
 $\therefore \frac{AE}{EF} = \frac{EF}{BE}$,
 $\therefore \frac{AE}{EF} = \frac{EF}{BE} = \frac{FD}{BC} = \frac{AD}{CF}$,
 $\therefore$ 矩形 $AEFD \sim$ 矩形 $EBCF$.

(2)【答案】证明见解析

【解析】 $\because$ 四边形 $ABCD$ 是梯形,

$AD \parallel BC$,

又 $\because EF \parallel AD$,

$\therefore EF \parallel AD \parallel BC$,

$\therefore \angle A = \angle BEF, \angle AEF = \angle B, \angle DFE = \angle C,$
 $\angle D = \angle EFC$,

$\therefore$ 梯形 $ABCD$ 中, $AB = CD$,

$\therefore$ 梯形 $ABCD$ 是等腰梯形,

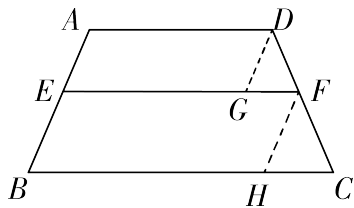
$\therefore \angle B = \angle C$,

$\therefore \angle AEF = \angle DFE$,

$\therefore$ 梯形 $AEFD$ 和 $EBCF$ 都是等腰梯形,

$\therefore AE = DF, EB = FC$,

如图,作 $DG \parallel AE$ 交 EF 于 G ,作 $FH \parallel EB$ 交 BC 于 H ,



则四边形 $AEGD$ 和 $EBHF$ 都是平行四边形,

$\therefore AE = DG = DF, EB = FH = FC$,

$AD = EG, EF = BH$,

$\angle DGF = \angle AEF, \angle FHC = \angle B$,

$\therefore \angle DGF = \angle FHC$,

$\therefore \triangle DGF \sim \triangle FHC$,

$\therefore \frac{DG}{FH} = \frac{GF}{HC} = \frac{DF}{FC}$,

$\therefore EF^2 = AD \cdot BC$,

$\therefore \frac{EF}{BC} = \frac{AD}{EF}$,

根据等比性质可得 $\frac{EF - AD}{BC - EF} = \frac{EF}{BC}$, 即

$$\frac{GF}{HC} = \frac{EF}{BC},$$

$$\therefore \frac{DF}{FC} = \frac{EF}{BC},$$

$$\therefore \frac{AE}{EB} = \frac{DF}{FC} = \frac{EF}{BC} = \frac{AD}{EF},$$

$\therefore$ 梯形 $AEFD \sim$ 梯形 $EBCF$.

4. (1)【答案】① 证明见解析;

② 成立,理由见解析.

【解析】① 证明: $\because CE \perp BF$,

$\therefore \angle BED = 90^\circ$,

$\therefore \angle DAC = 90^\circ$,

$\therefore \angle BED = \angle CAD = 90^\circ$,

$\therefore \angle BDE = \angle ADC$,

$\therefore \angle ABF = \angle ACD$,

$\therefore \angle BAC = 90^\circ, \angle ACB = 45^\circ$,

$\therefore \angle ABC = \angle ACB = 45^\circ$,

$\therefore AB = AC$,

在 $\triangle BAF$ 和 $\triangle CAD$ 中,

$$\begin{cases} \angle ABF = \angle ACD \\ AB = AC \\ \angle BAF = \angle CAD = 90^\circ \end{cases},$$

$\therefore \triangle BAF \cong \triangle CAD (ASA)$,

$\therefore BF = CD$,

$\therefore \angle BCP = \frac{1}{2} \angle ACB$,

$\therefore \angle BCE = \angle ACE = 22.5^\circ$,

$\therefore \angle CBE = \angle CFE = 67.5^\circ$,

$\therefore CB = CF$,

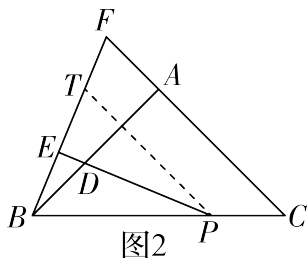
$\therefore CE \perp BF$,

$\therefore BE = EF$,

$\therefore BE = \frac{1}{2} CD$, 即 $BE = \frac{1}{2} PD$;

② 结论 $BE = \frac{1}{2} PD$ 成立.

理由:如图2中,过点P作 $PT//CF$ 交BF于点T.



$\because PT//CF$,

$\therefore \angle BPT = \angle C$,

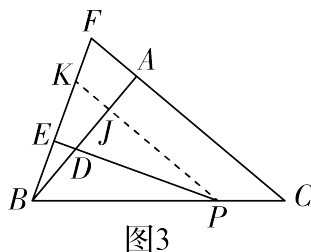
$\because \angle BPD = \frac{1}{2}\angle ACB$,

$\therefore \angle BPD = \frac{1}{2}\angle BPT$,

由(1)可知, $BE = \frac{1}{2}PD$.

(2)【答案】 $\frac{BE}{PD} = \frac{n}{2}$,过程见解析.

【解析】如图3,过点P作 $PK//CA$ 交BF于点K,交AB于点J.



$\because PK//AC$,

$\therefore \frac{PJ}{AC} = \frac{BJ}{BA}$,

$\therefore \frac{BJ}{PJ} = \frac{BA}{AC} = n$,

同法可证 $PB = PK$, $BE = EK$,

$\because \angle KBJ = \angle DPJ$, $\angle BJK = \angle PJD = 90^\circ$,

$\therefore \triangle BJK \sim \triangle PJD$,

$\therefore \frac{BK}{PD} = \frac{BJ}{PJ} = n$,

$\therefore \frac{2BE}{PD} = n$,

$\therefore \frac{BE}{PD} = \frac{n}{2}$.

第11讲 圆与多边形综合

1. (1)【答案】证明见解析.

【解析】 $\because M, N$ 分别是 AB, AC 的中点,

OM, ON 过圆心,

$\therefore OM \perp AB, ON \perp AC$.

又 $\because AB = AC$,

$\therefore AM = AN$.

$\because$ 在 $\text{Rt}\triangle AOM$ 和 $\text{Rt}\triangle AON$ 中,

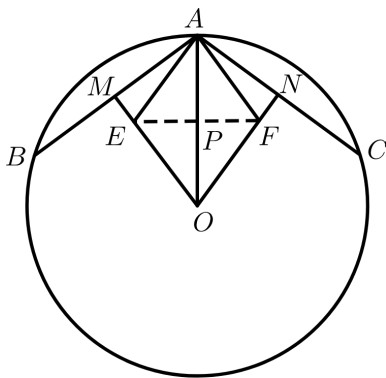
$$\begin{cases} AM = AN \\ OA = OA \end{cases},$$

$\therefore \text{Rt}\triangle AOM \cong \text{Rt}\triangle AON (\text{HL})$,

$\therefore \angle AOM = \angle AON$.

(2)【答案】证明见解析.

【解析】连接 EF , 交 AO 于点 P ,



$\because AE//ON, AF//OM$,

$\therefore$ 四边形 $AEOF$ 是平行四边形.

$\because AE//ON$,

$\therefore \angle EAO = \angle AON$.

$\because \angle AOM = \angle AON$,

$\therefore \angle AOM = \angle EAO$.

$\therefore AE = EO$.

$\therefore$ 四边形 $AEOF$ 是菱形.

$\therefore EF \perp AO, PO = \frac{1}{2}AO$.

$\because OM \perp AB$,

$\therefore \angle EPO = \angle AMO = 90^\circ$.

$\because \angle AOM = \angle AOM$,

$\therefore \triangle EPO \sim \triangle AMO$.

$\therefore \frac{OE}{AO} = \frac{OP}{OM}$.

$\therefore OE \cdot OM = AO \cdot OP = \frac{1}{2}AO^2$,

即 $OE \cdot OM = \frac{1}{2}AO^2$.

2. (1) 【答案】见解析

【解析】当点 N 与点 B 重合时, 以点 O 为圆心, $\frac{3}{2}$ 为半径的圆与直线 AP 相交. 理由如下:

作 $OG \perp AB$ 于 G , 如图 2 所示:

$\because \angle OBG = \angle ABM, \angle OGB = \angle AMB$,

$\therefore \triangle OBG \sim \triangle ABM$,

$\therefore \frac{BM}{AB} = \frac{BG}{OB}$, 即 $\frac{BM}{8} = \frac{4}{5}$,

解得: $BM = \frac{32}{5}$,

$\therefore OM = \frac{32}{5} - 5 = \frac{7}{5}$,

$\because \frac{7}{5} < \frac{3}{2}$,

$\therefore$ 当点 N 与点 B 重合时, 以点 O 为圆心, $\frac{3}{2}$ 为半径的圆与直线 AP 相交.

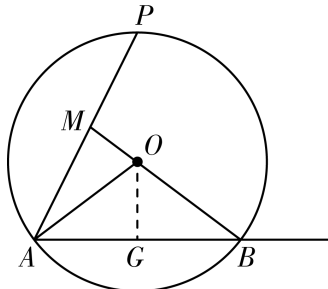


图2

(2) 【答案】见解析

【解析】① 当点 N 在线段 AB 延长线上时,

当圆 N 与圆 O 相外切时, 作 $OD \perp AB$ 于 D , 如图 3 所示.

$\because OA = OB = 5$,

$\therefore AD = DB = \frac{1}{2}AB = 4$,

$\therefore OD = \sqrt{OB^2 - BD^2} = \sqrt{5^2 - 4^2} = 3$,

$\because \angle BNO = \angle BON$,

$\therefore BN = OB = 5$,

$\therefore DN = DB + BN = 9$,

在 $Rt\triangle ODN$ 中, 由勾股定理得:

$ON = \sqrt{OD^2 + DN^2} = \sqrt{3^2 + 9^2} = 3\sqrt{10}$,

$\because$ 圆 N 与圆 O 相切,

$\therefore$ 圆 N 半径 $= ON - 5 = 3\sqrt{10} - 5$;

当圆 N 与圆 O 相内切时, 圆 N 半径 $= ON + 5 = 3\sqrt{10} + 5$;

② 当点 N 在线段 AB 上时, 此时点 P 在弦 AB 的下方, 点 N 在圆 O 内部, 只存在圆 N 与圆 O 相内切, 如图 4 所示,

作 $OE \perp AB$ 于 E , 则 $AE = BE = 4$,
 $OE = \sqrt{OB^2 - BE^2} = 3$,

$\because \angle BNO = \angle BON$,

$\therefore BN = OB = 5$,

$\therefore EN = BN - BE = 1$,

在 $Rt\triangle OEN$ 中, 由勾股定理得:

$ON = \sqrt{OE^2 + EN^2} = \sqrt{3^2 + 1^2} = \sqrt{10}$,

$\therefore$ 圆 N 半径 $= 5 - ON = 5 - \sqrt{10}$ 或圆 N 半径 $= 5 + ON = 5 + \sqrt{10}$;

综上所述, 当 $\angle BNO = \angle BON$, 且圆 N 与圆 O 相切时, 圆 N 半径的长为 $3\sqrt{10} - 5$ 或 $3\sqrt{10} + 5$ 或 $5 - \sqrt{10}$ 或 $5 + \sqrt{10}$.

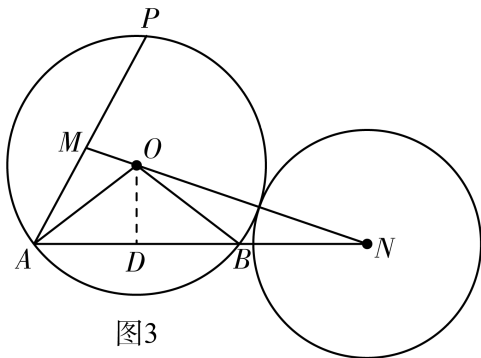


图3

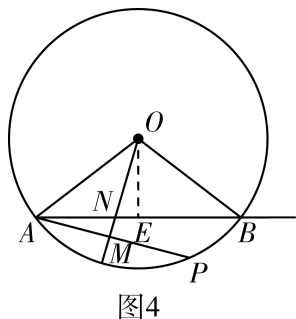
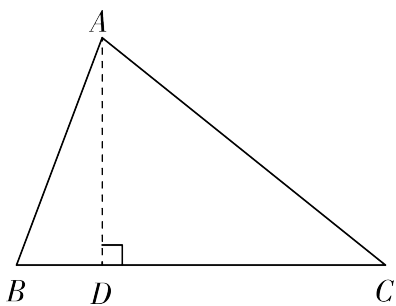


图4

3. (1) 【答案】 $\frac{4\sqrt{2}}{9}$.

【解析】过点 A 作 $AD \perp BC$ 于点 D, 如图:



$$\because \cos \angle ABC = \frac{1}{3},$$

$$\therefore BD = \frac{1}{3}AB = 2,$$

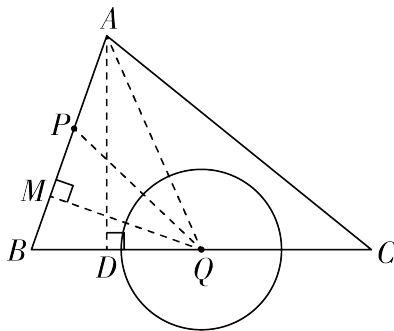
$$\therefore CD = BC - BD = 7, AD = \sqrt{AB^2 - BD^2} = 4\sqrt{2},$$

$$\therefore AC = \sqrt{AD^2 + CD^2} = 9,$$

$$\therefore \sin C = \frac{AD}{AC} = \frac{4\sqrt{2}}{9}.$$

(2) 【答案】 $y = \frac{-2x + \sqrt{4x^2 - 36x + 153}}{2} \left(1 \leq x < \frac{17}{4} \right).$

【解析】连接 PQ, AQ , 过点 Q 作 $QM \perp AB$ 于点 M , 如图:



$\because Q$ 是 BC 中点,

$$\therefore BQ = \frac{9}{2},$$

$$\therefore BD = 2,$$

$$\therefore DQ = \frac{5}{2},$$

$$\therefore AQ = \sqrt{AD^2 + DQ^2} = \frac{3}{2}\sqrt{17},$$

$$\because \cos \angle ABC = \frac{1}{3},$$

$$\therefore BM = \frac{1}{3}BQ = \frac{3}{2},$$

$$\therefore AM = \frac{9}{2},$$

$$\therefore MQ = \sqrt{AQ^2 - AM^2} = 3\sqrt{2},$$

$\because \odot P$ 与 $\odot Q$ 外切, $\odot P$ 经过点 A,

$$\therefore PQ = x + y, AP = x,$$

当 P 在 AM 上时, $PM = \frac{9}{2} - x$, 当 P 在 BM 上时, $PM = x - \frac{9}{2}$,

在 $\text{Rt}\triangle PMQ$ 中, $PQ^2 = PM^2 + MQ^2$,

$$\text{即 } (x+y)^2 = \left(x - \frac{9}{2}\right)^2 + 18,$$

$$\text{整理得: } y^2 + 2xy + 9x - \frac{153}{4} = 0,$$

$$\because x > 0, y > 0,$$

$$\therefore y = \frac{-2x + \sqrt{4x^2 - 36x + 153}}{2} \left(0 < x < \frac{17}{4} \right);$$

$\because \odot Q$ 的直径不大于 BC ,

$$\therefore y \leq \frac{9}{2},$$

$$\therefore \frac{-2x + \sqrt{4x^2 - 36x + 153}}{2} \leq \frac{9}{2},$$

$$\therefore x \geq 1,$$

$$\therefore y = \frac{-2x + \sqrt{4x^2 - 36x + 153}}{2} \left(1 \leq x < \frac{17}{4} \right).$$

(3) 【答案】AP 的长为 3 或 $\frac{3}{2}$.

【解析】 $\because \triangle BPQ$ 是等腰三角形,

当 $PQ = BQ$ 时, $x + y = \frac{9}{2}$,

$$\text{即 } \frac{-2x + \sqrt{4x^2 - 36x + 153}}{2} + x = \frac{9}{2},$$

解得: $x = 3$ 或 6 (不在定义域内, 舍去),

当 $BQ = BP$ 时, $\frac{9}{2} = 6 - x$,

$$\text{解得: } x = \frac{3}{2},$$

当 $PQ = BP$ 时,

$$6 - x = \frac{-2x + \sqrt{4x^2 - 36x + 153}}{2},$$

解得 x 无实数根,

综上所述, AP 的长为 3 或 $\frac{3}{2}$.

第 13 讲 二次函数与几何综合 (一)

1. (1) 【答案】 $y = -x^2 + 2x + 3$.

【解析】 $y = -x + n$ 过点 $A(3, 0)$,

$$\therefore 0 = -3 + n,$$

$$\therefore n = 3,$$

$$\therefore y = -x + 3,$$

当 $x = 0$ 时, $y = 3$,

$$\therefore B(0, 3).$$

$\because y = -x^2 + bx + c$ 经过点 A, B ,

$$\therefore \begin{cases} 0 = -9 + 3b + c \\ 3 = c \end{cases},$$

$$\therefore \begin{cases} b = 2 \\ c = 3 \end{cases},$$

$\therefore$ 抛物线解析式为 $y = -x^2 + 2x + 3$.

(2) 【答案】① $(2, 0)$ 或 $(1, 0)$.

② $m = \frac{7}{3}$ 或 5 .

【解析】① 当 $\angle BPD = 90^\circ$ 时,

点 P 纵坐标与点 B 相同,

$$\therefore y = 3 \text{ 时,}$$

$$3 = -x^2 + 2x + 3,$$

解得 $x = 0$ (舍) 或 $x = 2$,

$$\therefore P(2, 3),$$

$$\therefore E(2, 0),$$

当 $\angle PBD = 90^\circ$ 时,

$$\therefore BP \perp BA,$$

$\therefore$ 设直线 BP 的解析式为 $y = kx + a$,

$$\therefore k = 1, \text{ 经过点 } B(0, 3),$$

$$\therefore a = 3,$$

$$\therefore \text{直线 } BP: y = x + 3,$$

$$\text{联立 } \begin{cases} y = x + 3 \\ y = -x^2 + 2x + 3 \end{cases},$$

$$\therefore x + 3 = -x^2 + 2x + 3,$$

$$x^2 - x = 0,$$

$$\therefore x = 0 \text{ (舍) 或 } x = 1,$$

$$\therefore P(1, 4),$$

$$\therefore E(1, 0).$$

综上 E 的坐标为 $(2, 0)$ 或 $(1, 0)$.

② $\because OB = OA = 3$,

$$\therefore \angle OBA = 45^\circ,$$

当 E 在 OA 上时,

P 在 BA 上方,

若 $\angle PBD + \angle CBO = 45^\circ$,

则 $\angle PBD + \angle CBO + \angle OBA = 90^\circ$,

$\therefore BP \perp BC$,

$\because B(0,3), C(-1,0)$,

$\therefore$ 直线 BC 解析式为 $y = 3x + 3$,

$\therefore$ 直线 BP 解析式为 $y = -\frac{1}{3}x + 3$,

$$\text{联立} \begin{cases} y = -\frac{1}{3}x + 3 \\ y = -x^2 + 2x + 3 \end{cases},$$

解得 $x = 0$ (舍) 或 $x = \frac{7}{3}$,

$\therefore m = \frac{7}{3}$,

当 E 在 A 点右侧,

即 P 点在 BA 下方,

若 $\angle PBD + \angle CBO = 45^\circ$,

$\because \angle PBO + \angle PBD = 45^\circ$,

$\therefore \angle PBO = \angle CBO$,

$\therefore$ 点 C 关于 y 轴的对称点 $(1,0)$,

$\therefore$ 直线 PB 解析式为 $y = -3x + 3$,

$$\text{联立} \begin{cases} y = -3x + 3 \\ y = -x^2 + 2x + 3 \end{cases},$$

解得: $x = 0$ (舍) 或 $x = 5$,

$\therefore m = 5$,

综上 $m = \frac{7}{3}$ 或 5 .

2. (1) 【答案】 $y = -x^2 + 2x + 3, (1,4)$.

【解析】由题意, 可得
$$\begin{cases} 0 = a - b + 3 \\ 1 = -\frac{b}{2a} \end{cases} \Rightarrow$$

$$\begin{cases} a = -1 \\ b = 2 \end{cases},$$

$\therefore$ 抛物线解析式为 $y = -x^2 + 2x + 3$,

$\therefore$ 顶点 D 的坐标为 $(1,4)$.

(2) 【答案】 $P(-2, -5)$.

【解析】设 $P(t, -t^2 + 2t + 3)$,

$\because P$ 在第三象限,

$\therefore t < -1$,

由抛物线解析式易求 $C(0,3), B(3,0)$,

根据四边形 $PBDC$ 为梯形, 可分为两种情况:

① $PB \parallel CD$,

$$k_{PB} = \frac{-t^2 + 2t + 3 - 0}{t - 3},$$

$$k_{CD} = \frac{4 - 3}{1 - 0} = 1,$$

$$\therefore \frac{-t^2 + 2t + 3}{t - 3} = 1,$$

解得 $t = 3$ 或 -2 ,

$\because t < -1$,

$\therefore t = -2$,

$\therefore P(-2, -5)$.

② $PC \parallel DB$,

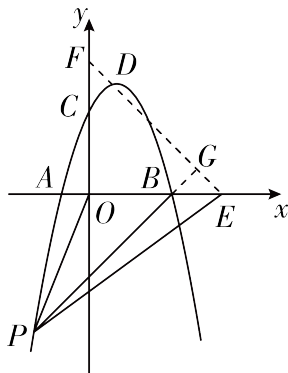
$$\text{同理可得 } \frac{-t^2 + 2t + 3 - 3}{t - 0} = \frac{4 - 0}{1 - 3},$$

解得 $t = 0$ 或 $t = 4$, 均不符合题意.

综上所述: 点 P 的坐标为 $(-2, -5)$.

(3) 【答案】 $\frac{29}{3}$.

【解析】在 y 轴正半轴上找一点 F , 使 $OF = OE$, 连接 EF , 延长 PB 交 EF 于 G 点,



由(2)知 $P(-2, -5), B(3, 0)$,

$$\therefore l_{PB}: y = x - 3,$$

$\therefore$ 直线 PB 与 y 轴交点为 $(0, -3)$,

$$\therefore \angle PBO = 45^\circ,$$

设 $OE = OF = b (b > 0), E(b, 0), F(0, b)$,

易求 $l_{EF}: y = -x + b$,

$$\because k_{PB} \cdot k_{EF} = -1,$$

$$\therefore PB \perp EF,$$

$$\text{联立方程 } \begin{cases} y = x - 3 \\ y = -x + b \end{cases} \Rightarrow \begin{cases} x = \frac{b+3}{2} \\ y = \frac{b-3}{2} \end{cases},$$

$$\therefore G\left(\frac{b+3}{2}, \frac{b-3}{2}\right),$$

$$\because OF = OE, \angle FOE = 90^\circ,$$

$$\therefore \angle FEO = 45^\circ,$$

$$\therefore \tan(\angle PBO + \angle PEO)$$

$$= \tan(45^\circ + \angle PEO)$$

$$= \tan(\angle FEO + \angle PEO)$$

$$= \tan \angle PEF$$

$$= \frac{5}{2},$$

$$\therefore \frac{PG}{GE} = \frac{\sqrt{\left(\frac{b+3}{2} + 2\right)^2 + \left(\frac{b-3}{2} + 5\right)^2}}{\sqrt{\left(\frac{b+3}{2} - b\right)^2 + \left(\frac{b-3}{2}\right)^2}} =$$

$$\frac{|b+7|}{|b-3|} = \frac{5}{2},$$

$b = \frac{29}{3}$ 或 $b = \frac{1}{7}$ (舍去, $\angle FEP$ 为钝角与图形不符),

$$\therefore OE \text{ 的长是 } \frac{29}{3}.$$

3. (1) 【答案】 $A(0, -4), B(2, -4)$.

【解析】 抛物线的对称轴为: 直线 $x =$

$$-\frac{b}{2a} = 1,$$

对于 $y = ax^2 - 2ax - 4$, 令 $x = 0$, 则 $y = -4$,

即点 $A(0, -4)$,

根据抛物线的对称性, 则点 $B(2, -4)$,

即点 A, B 的坐标分别为: $(0, -4), (2, -4)$.

$$(2) \text{【答案】 } ① y = \frac{1}{12}x^2 - \frac{1}{6}x - 4;$$

$$② \text{ 存在, } P \text{ 点坐标为: } \left(1, -\frac{15}{2}\right) \text{ 或 } \left(1, \frac{1}{2}\right).$$

【解析】 ① 由点 B 的坐标得, $\tan \angle AOB =$

$$\frac{AB}{OA} = \frac{1}{2},$$

$$\because AC \perp OB,$$

$$\therefore \tan \angle OAC = 2,$$

$$\because OA = 4,$$

$$\therefore OC = 8, \text{ 即点 } C(8, 0),$$

将点 C 的坐标代入抛物线表达式得: $64a -$

$$16a - 4 = 0,$$

$$\text{解得: } a = \frac{1}{12},$$

$$\text{则抛物线的表达式为: } y = \frac{1}{12}x^2 - \frac{1}{6}x - 4.$$

② 存在, 理由如下:

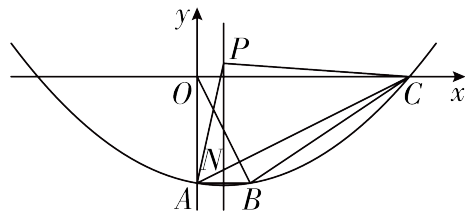
$$\because A(0, -4), B(2, -4),$$

$$\therefore AB = 2,$$

$$\therefore S_{\triangle ABC} = \frac{1}{2} \times 2 \times 4 = 4,$$

设点 P 的坐标为 $(1, m)$,

如图, 抛物线对称轴交 AC 于 N 点,



$$\because A(0, -4), C(8, 0),$$

$\therefore$ 利用待定系数法可求出直线 AC 的解析

$$\text{式为: } y = \frac{1}{2}x - 4,$$

$$\text{即 } x = 1 \text{ 时, } y = \frac{1}{2}x - 4 = -\frac{7}{2},$$

即 $N\left(1, -\frac{7}{2}\right)$,

则有: $PN = \left|m + \frac{7}{2}\right|$,

$$\therefore S_{\triangle PAC} = \frac{1}{2} \times PN \times x_C = \frac{1}{2} \times \left|m + \frac{7}{2}\right| \times 8,$$

$$\text{即有: } \frac{1}{2} \times \left|m + \frac{7}{2}\right| \times 8 = 4 \times 4 = 16,$$

$$\text{解得 } m = -\frac{15}{2} \text{ 或 } m = \frac{1}{2},$$

$\therefore$ 点 P 的标为 $\left(1, -\frac{15}{2}\right)$ 或 $\left(1, \frac{1}{2}\right)$.

第 15 讲 二次函数与几何综合 (二)

1. (1) 【答案】(2, -9)

【解析】 $\because$ 抛物线 $y = x^2 + bx + c$ 经过点

$A(-2, 7)$, 点 $C(5, 0)$,

$$\therefore \begin{cases} 4 - 2b + c = 7 \\ 25 + 5b + c = 0 \end{cases},$$

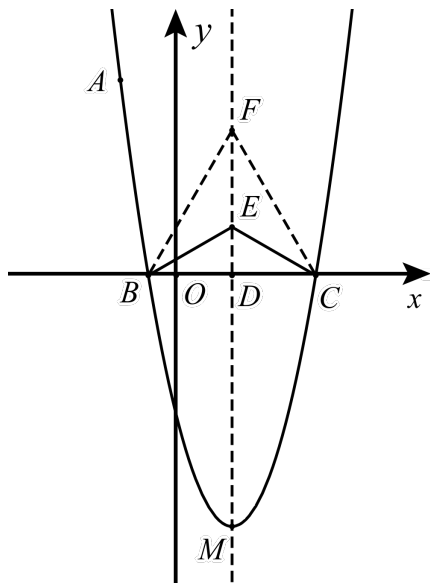
$$\text{解得: } \begin{cases} b = -4 \\ c = -5 \end{cases},$$

$$\therefore y = x^2 - 4x - 5 = (x - 2)^2 - 9,$$

$\therefore$ 抛物线的顶点 M 的坐标为 (2, -9).

(2) 【答案】 $E(2, \sqrt{3})$

【解析】设 $E(2, t)$, 且 $t > 0$, 如图, 连接 CF , 设抛物线的对称轴交 x 轴于 D ,



则 $D(2, 0)$,

$\because$ 点 B, C 关于直线 $x = 2$ 对称,

$$\therefore B(-1, 0),$$

$$\therefore BC = 6, BD = CD = 3, DE = t,$$

由翻折得: $BF = BC = 6$, $\angle FBE = \angle CBE = \frac{1}{2} \angle CBF$,

$\because$ 点 F 在抛物线的对称轴上,

$$\therefore CF = BF = 6,$$

$$\therefore CF = BF = BC,$$

$\therefore \triangle BCF$ 是等边三角形,

$$\therefore \angle CBF = 60^\circ,$$

$$\therefore \angle CBE = 30^\circ,$$

$$\therefore \frac{DE}{BD} = \tan \angle CBE = \tan 30^\circ = \frac{\sqrt{3}}{3},$$

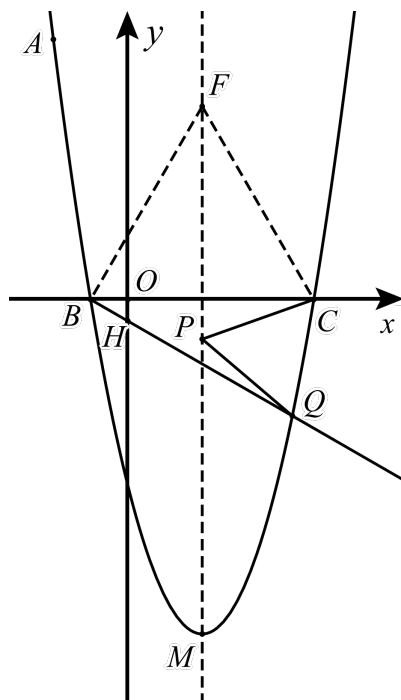
$$\therefore \frac{t}{3} = \frac{\sqrt{3}}{3},$$

$$\therefore t = \sqrt{3},$$

$$\therefore E(2, \sqrt{3}).$$

(3) 【答案】 $y = -\frac{\sqrt{3}}{3}x - \frac{\sqrt{3}}{3}$

【解析】取 (2) 中的点 F , 连接 BF, CF , 设直线 BQ 交 y 轴于点 H ,



$$\because BF = BC = CF = 6,$$

$\therefore \triangle FCB$ 为等边三角形,

$$\therefore \angle BCF = \angle BFC = 60^\circ,$$

$$\because FM \perp BC,$$

$$\therefore \angle CFP = \frac{1}{2} \angle BFC = 30^\circ,$$

$\because$ 点 P 在抛物线的对称轴上, 点 Q 是抛物线上位于第四象限内的点, $\triangle CPQ$ 为等边三角形,

$$\therefore CP = CQ, \angle PCQ = 60^\circ,$$

$$\therefore \angle BCF + \angle BCP = \angle PCQ + \angle BCP, \quad \text{即} \\ \angle FCP = \angle BCQ,$$

$$\therefore \triangle CFP \cong \triangle CBQ (\text{SAS}),$$

$$\therefore \angle CBQ = \angle CFP = 30^\circ,$$

在 $\text{Rt}\triangle BOH$ 中, $OB = 1$, $\angle BOH = 90^\circ$, $\angle HBO = 30^\circ$,

$$\therefore \frac{OH}{OB} = \tan \angle HBO = \tan 30^\circ = \frac{\sqrt{3}}{3},$$

$$\therefore OH = \frac{\sqrt{3}}{3},$$

$$\therefore H \left(0, -\frac{\sqrt{3}}{3} \right),$$

设直线 BQ 的函数表达式为 $y = mx + n$, 把 $B(-1, 0)$ 、 $H \left(0, -\frac{\sqrt{3}}{3} \right)$ 代入, 得:

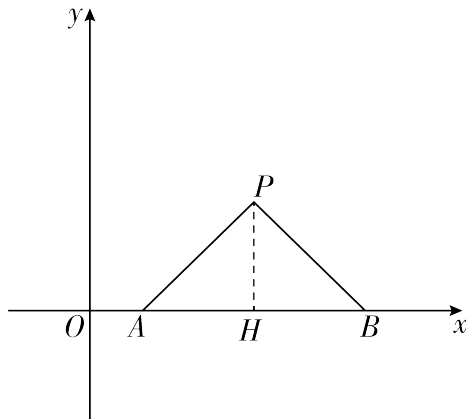
$$\begin{cases} -m + n = 0 \\ n = -\frac{\sqrt{3}}{3} \end{cases},$$

$$\text{解得: } \begin{cases} m = -\frac{\sqrt{3}}{3} \\ n = -\frac{\sqrt{3}}{3} \end{cases},$$

$$\therefore \text{直线 } BQ \text{ 的表达式为 } y = -\frac{\sqrt{3}}{3}x - \frac{\sqrt{3}}{3}.$$

$$2. (1) \text{【答案】} y = -\frac{1}{3}x^2 + \frac{8}{3}x - \frac{7}{3}.$$

【解析】过 P 作 $PH \perp x$ 轴于 H , 如图:



$\because P$ 为抛物线的顶点,

$$\therefore PA = PB,$$

$$\because \angle APB = 90^\circ, PH \perp x \text{ 轴},$$

$$\therefore AH = PH = BH,$$

$\because$ 抛物线的顶点 $P(4, 3)$,

$$\therefore OH = 4, PH = 3 = AH = BH, \text{ 且 } y =$$

$$a(x-4)^2 + 3,$$

$$\therefore OA = OH - AH = 4 - 3 = 1,$$

$$\therefore A(1, 0),$$

把 $A(1, 0)$ 代入 $y = a(x-4)^2 + 3$ 得:

$$0 = 9a + 3,$$

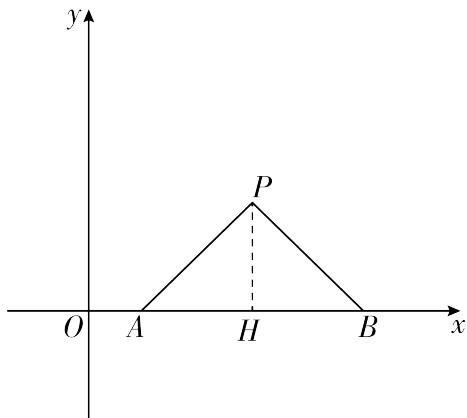
$$\text{解得: } a = -\frac{1}{3},$$

$$\therefore y = -\frac{1}{3}(x-4)^2 + 3 = -\frac{1}{3}x^2 + \frac{8}{3}x - \frac{7}{3},$$

$$\therefore \text{抛物线的表达式为 } y = -\frac{1}{3}x^2 + \frac{8}{3}x - \frac{7}{3}.$$

(2)【答案】 $an+1=0$.

【解析】过 P 作 $PH \perp x$ 轴于 H , 如图:



$\because P$ 为抛物线的顶点,

$\therefore PA = PB$,

$\because \angle APB = 90^\circ, PH \perp x$ 轴,

$\therefore AH = PH = BH$,

$\because$ 抛物线 $y = a(x-m)^2 + n$ 的顶点 P 坐标为 (m, n) ,

$\therefore AH = PH = BH = n, OH = m$,

$\therefore OA = m - n$,

$\therefore A$ 的坐标为 $(m-n, 0)$,

把 $A(m-n, 0)$ 代入 $y = a(x-m)^2 + n$ 得:

$0 = an^2 + n$,

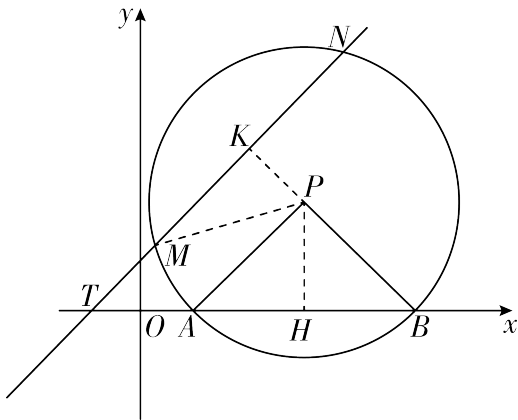
$\therefore P(m, n)$ 在第一象限,

$\therefore n \neq 0$,

$\therefore an+1=0$.

(3)【答案】 $-\frac{\sqrt{14}}{2a}$.

【解析】解: 延长 BP 交 MN 于 K , 连接 PM , 过 P 作 $PH \perp x$ 轴于 H , 设直线 MN 交 x 轴于 T , 如图:



由 $y = x + \frac{n}{2}$ 可知, $\angle KTB = 45^\circ = \angle PAB$,

$\therefore KT \parallel AP$,

$\therefore \angle TKB = \angle APB = 90^\circ$,

$\therefore \triangle KTB$ 是等腰直角三角形,

$\because P(m, n)$ 在直线 $y = \frac{1}{2}x$ 上,

$\therefore n = \frac{1}{2}m$,

$\therefore m = 2n$,

$\therefore P(2n, n)$,

同 (2) 可知, $AH = PH = BH = n$,

$OA = OH - AH = n, OB = OH + BH = 3n$,

$\therefore A(n, 0), B(3n, 0)$,

$\therefore BP = AP = \sqrt{AH^2 + PH^2} = \sqrt{2}n = PM$,

在 $y = x + \frac{n}{2}$ 中, 令 $y = 0$ 得 $x = -\frac{n}{2}$,

$\therefore T(-\frac{n}{2}, 0)$,

$\therefore BT = 3n - (-\frac{n}{2}) = \frac{7}{2}n$,

$\therefore BK = \frac{BT}{\sqrt{2}} = \frac{7\sqrt{2}}{4}n$,

$\therefore PK = BK - BP = \frac{7\sqrt{2}}{4}n - \sqrt{2}n = \frac{3\sqrt{2}}{4}n$,

$\therefore MK = \sqrt{PM^2 - PK^2} = \sqrt{(\sqrt{2}n)^2 - \left(\frac{3\sqrt{2}}{4}n\right)^2} = \frac{\sqrt{14}}{4}n$,

由垂径定理可知, $MN = 2MK = \frac{\sqrt{14}}{2}n$,

由(2)知 $an+1=0$,

$$\therefore n = -\frac{1}{a},$$

$$\therefore MN = \frac{\sqrt{14}}{2} \times \left(-\frac{1}{a}\right) = -\frac{\sqrt{14}}{2a};$$

$$\therefore MN \text{ 的长为 } -\frac{\sqrt{14}}{2a}.$$

3. (1) 【答案】 $y = -\frac{3}{4}x^2 + \frac{9}{4}x + 3$

【解析】 $\because$ 抛物线 $y = -\frac{3}{4}x^2 + bx + c$ 与 x 轴

交于点 $A(4,0)$, 与 y 轴交于点 $C(0,3)$,

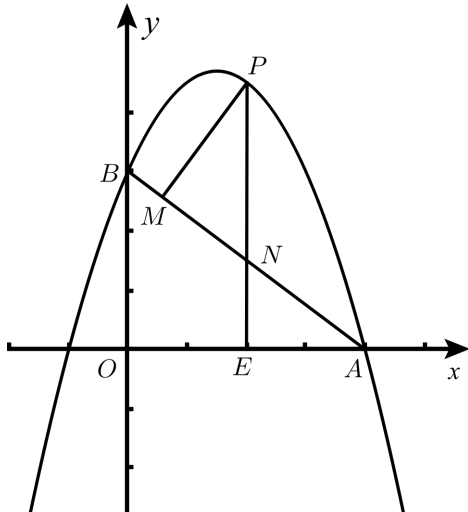
$$\therefore \begin{cases} -12 + 4b + c = 0 \\ 3 = c \end{cases}$$

$$\therefore \begin{cases} b = \frac{9}{4} \\ c = 3 \end{cases},$$

$$\therefore y = -\frac{3}{4}x^2 + \frac{9}{4}x + 3.$$

(2) 【答案】 $\left(2, \frac{9}{2}\right)$

【解析】 $\because PM \perp AB, PE \perp x$ 轴,



$$\therefore \angle PMN = \angle PEA = 90^\circ,$$

$$\text{又 } \angle PNM = \angle ANE,$$

$$\therefore \triangle PMN \sim \triangle AEN,$$

$$\therefore \frac{C_{\triangle PMN}}{C_{\triangle AEN}} = \frac{PN}{AN}. \text{ 即 } \frac{C_1}{C_2} = \frac{PN}{AN}.$$

$$\text{又 } \frac{C_1}{C_2} = \frac{6}{5},$$

$$\therefore \frac{PN}{AN} = \frac{6}{5},$$

设直线 $AB: y = kx + b$, 又直线 AB 经过点

$A(4,0)$, 点 $B(0,3)$,

$$\therefore \begin{cases} 4k + b = 0 \\ 3 = b \end{cases},$$

$$\therefore \begin{cases} k = -\frac{3}{4} \\ b = 3 \end{cases},$$

$$\therefore y = -\frac{3}{4}x + 3,$$

$\because$ 点 P 在抛物线 $y = -\frac{3}{4}x^2 + \frac{9}{4}x + 3$ 上,

$$\therefore \text{设点 } P\left(m, -\frac{3}{4}m^2 + \frac{9}{4}m + 3\right),$$

$\because$ 点 N 在直线 $y = -\frac{3}{4}x + 3$ 上,

$$\text{设点 } N\left(m, -\frac{3}{4}m + 3\right),$$

$$\therefore PN = \left(-\frac{3}{4}m^2 + \frac{9}{4}m + 3\right) - \left(-\frac{3}{4}m + 3\right) = -\frac{3}{4}m^2 + 3m,$$

$$\text{又 } AN = \sqrt{(m-4)^2 + \left(-\frac{3}{4}m + 3\right)^2} = \frac{5}{4}(4-m),$$

$$\therefore \frac{-\frac{3}{4}m^2 + 3m}{\frac{5}{4}(4-m)} = \frac{6}{5}.$$

解得 $m_1 = 2, m_2 = 4$ (不合题意, 舍去),

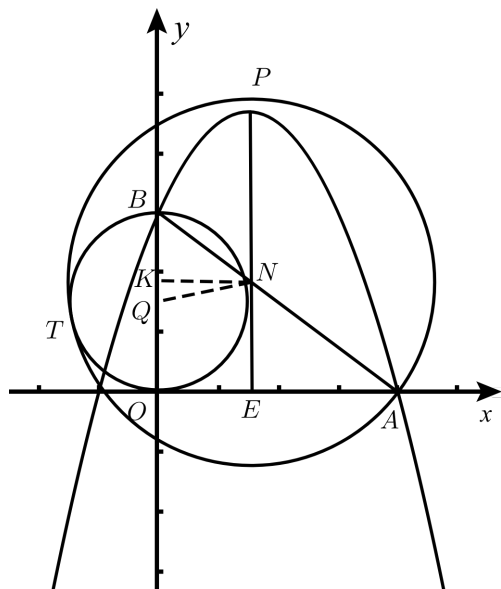
$$\therefore \text{点 } P \text{ 的坐标是 } \left(2, \frac{9}{2}\right).$$

(3) 【答案】 $\frac{20}{13}$

【解析】设 OB 的中点为点 Q , 则点 Q 的坐标为 $\left(0, \frac{3}{2}\right)$,

$$\text{又点 } N\left(m, -\frac{3}{4}m + 3\right),$$

过点 N 作 $NK \perp y$ 轴于点 K ,



则 $NK = m$, $KQ = -\frac{3}{4}m + 3 - \frac{3}{2} = -\frac{3}{4}m + \frac{3}{2}$,

在 $\text{Rt}\triangle NQK$ 中,

$$\therefore QN = \sqrt{NK^2 + KQ^2}$$

$$= \sqrt{m^2 + \left(-\frac{3}{4}m + \frac{3}{2}\right)^2}$$

$$= \sqrt{\frac{25}{16}m^2 - \frac{9}{4}m + \frac{9}{4}},$$

当 $\odot N$ 与 $\odot Q$ 内切时, $QN = \left|AN - \frac{1}{2}OB\right|$,

$$\therefore \sqrt{\frac{25}{16}m^2 - \frac{9}{4}m + \frac{9}{4}} = \frac{5}{4}(4-m) - \frac{3}{2},$$

解得: $m = \frac{20}{13}$,

$\therefore$ 当 $\odot N$ 与 $\odot Q$ 内切时, $m = \frac{20}{13}$.